A SIGNED EXCEDANCE ENUMERATOR OF DERANGEMENTS

SHING HO J. LIN

ABSTRACT. The classical identity $E_n - O_n = (-1)^{n-1}(n-1)$, for the numbers of even and odd derangements of $[n]$, admits two refinements: one by fixed points and one by excedances. We show that both are marginals of a single Toeplitz determinant,

$$\sum_{\pi \in S_n} \operatorname{sgn}(\pi)\, y^{\operatorname{fix}(\pi)}\, q^{\operatorname{exc}(\pi)} = \frac{(y-q)^n - q(y-1)^n}{1-q},$$

specializing to the fixed-point identity studied by Pain and to the signed Eulerian identity of Désarménien and Foata. Its $y = 0$ fibre is the Mantaci–Rakotondrajao theorem $\sum_{\pi \in \mathcal{D}_n} \operatorname{sgn}(\pi)\, q^{\operatorname{exc}(\pi)} = (-1)^{n-1}(q + \cdots + q^{n-1})$: every excedance level of $\mathcal{D}_n$ is sign-balanced up to exactly one derangement. We record the exact sign-randomization principle that this yields, jointly for the pair (fix, exc); the $q = -1$ evaluation together with its bijective counterparts; and the infusion of the fixed-point weight into the catalog of entry-wise statistics of Sivasubramanian. In the hyperoctahedral group B_n the story is character-sensitive. The standard sign character gives an identically vanishing full-group sum and the constant derangement fibre $(-1)^n$; its twist by negation parity — which we identify with the sign of the underlying permutation — gives $2(1-q)^{n-1}$ and the twisted fibre $(-1)^{n-1}(1 + 2q + \cdots + 2q^{n-1})$, refined four-fold by the negation count modulo 4. The same graded philosophy is carried out for the Alon–Tarsi constant: graded by sparsity, its layers vanish identically at odd orders and are divisible by $n!$ at even orders, where the nonzero layers carry the sign $(-1)^k$. Finally, the signed excedance levels generate a subalgebra A_n of $\mathbb{Q}S_n$ which contains the group algebra of the long cycle, is semisimple for every n, and on which the augmentation and the sign are the only characters that extend to $\mathbb{Q}S_n$: the signed and the unsigned derangement enumerators are two eigenvalues of one integral operator. A cyclotomic family of further candidates, $\chi_a(\varepsilon_{n,k}) = \zeta_n^{ak}$, is proved to consist of characters for $n = 3, 5$ and proved not to for $n = 4, 6$.

1. INTRODUCTION

For $n \geq 1$ let $[n] = \{1, \ldots, n\}$, let S_n act on $[n]$ in one-line notation, and let

$$\mathcal{D}_n = \{\pi \in S_n : \pi(i) \neq i \text{ for all } i\}$$

be the set of *derangements*. Writing E_n and O_n for the numbers of even and odd derangements, the classical identity

$$E_n - O_n = (-1)^{n-1}(n-1) \tag{1}$$

has been in the literature at least since the problem sections of the 1950s; [3] collects references and gives a bijective proof. The shortest determinant proof observes that the all-ones matrix minus the identity satisfies

$$\det(J_n - I_n) = \sum_{\pi \in S_n} \operatorname{sgn}(\pi) \prod_i (J_n - I_n)_{i,\pi(i)} = \sum_{\pi \in \mathcal{D}_n} \operatorname{sgn}(\pi),$$

Date: October 29, 2026.

2020 *Mathematics Subject Classification.* Primary 05A05; Secondary 05A15, 05A19.

Key words and phrases. Derangements, excedances, sign-balance, signed Eulerian numbers, Toeplitz determinants.

because a product $\prod_i (J-I)_{i,\pi(i)}$ vanishes unless π has no fixed point, while the eigenvalues of $J_n - I_n$ are $n-1$ and -1 with multiplicity $n-1$, giving $\det(J_n - I_n) = (-1)^{n-1}(n-1)$. Benjamin, Bennett and Newberger [3] found a bijective proof of (1) via a sign-reversing involution. The refinement by fixed points is classical — it is the determinant of $J_n + (y-1)I_n$, i.e. the signed generating function of the rencontres numbers — and was taken up recently by Pain [13]:

$$\sum_{\pi \in S_n} \operatorname{sgn}(\pi)\, y^{\operatorname{fix}(\pi)} = \det\big(J_n + (y-1)I_n\big) = (y-1)^{n-1}(y+n-1), \tag{2}$$

whose coefficients are signed rencontres numbers (Pain's "odd–even staggering"; at $y=0$ this is (1)). On the other hand, Désarménien and Foata [6] initiated the study of *signed Eulerian numbers*; their identity reads

$$\sum_{\pi \in S_n} \operatorname{sgn}(\pi)\, q^{\operatorname{exc}(\pi)} = (1-q)^{n-1}, \qquad \operatorname{exc}(\pi) := \#\{i : \pi(i) > i\}. \tag{3}$$

Both refinements (2) and (3) are, in fact, the two marginals of a single statement one level up (Theorem 1): a two-variable identity, given by a Toeplitz determinant, whose left-hand side carries $\operatorname{sgn}(\pi)$, $y^{\operatorname{fix}(\pi)}$ and $q^{\operatorname{exc}(\pi)}$ simultaneously. Its $y=0$ fiber yields the refinement of (1) by excedances

$$F_n(q) := \sum_{\pi \in \mathcal{D}_n} \operatorname{sgn}(\pi)\, q^{\operatorname{exc}(\pi)} = (-1)^{n-1}\big(q + q^2 + \cdots + q^{n-1}\big), \tag{4}$$

so that *within every excedance level*, the even derangements outnumber the odd ones — or vice versa — by exactly one. We credit this statement honestly: (4) is a theorem of Mantaci–Rakotondrajao [9, Prop. 4.3], with a sign-reversing-involution proof by Ksavrelof–Zeng [8] and a determinant proof by Sivasubramanian [15], whose framework of signed excedance-like enumeration via determinants overlaps substantially with the present section (the matrix he attaches to $\operatorname{sgn}\cdot q^{\operatorname{exc}}$ is precisely the staircase determinant behind (3), and zeroing its diagonal is the specialization $y=0$). The two-variable identity (5) is likewise not new: writing $P_n(x,y,\beta) = \sum_{\sigma\in S_n} x^{\operatorname{exc}\sigma} y^{\operatorname{fix}\sigma} \beta^{\operatorname{cyc}\sigma}$, Ksavrelof–Zeng evaluate $P_n(x,y,-1) = \big((x-y)^n - x(1-y)^n\big)/(1-x)$ [8, (3)], which is (5) written with the cycle-parity sign $(-1)^{\operatorname{cyc}\pi} = (-1)^n \operatorname{sgn}(\pi)$. What the present note adds is therefore not the identity but its Toeplitz-determinant proof — the source of the recurrence (6) and hence of the $q=-1$ analysis of Section 3 — together with its consequences: the unification with (2) noted above, the joint sign-randomization principle of Corollary 6, and the type-B analogue of Section 4.

Theorem A (Mantaci–Rakotondrajao [9]; the $y=0$ fibre of Theorem 1 below, restated combinatorially as Corollary 2)**.** *For all $n \geq 2$ and every $1 \leq k \leq n-1$,*

$$\#\{\pi \in \mathcal{D}_n : \operatorname{exc}(\pi) = k,\ \pi \text{ even}\} - \#\{\pi \in \mathcal{D}_n : \operatorname{exc}(\pi) = k,\ \pi \text{ odd}\} = (-1)^{n-1}.$$

For instance, the even/odd pairs per excedance level k are $(1,0), (11,10), (11,10), (1,0)$ for $n=5$ and $(0,1), (25,26), (80,81), (25,26), (0,1)$ for $n=6$. Summing over k recovers (1); see Corollary 2 and Corollary 3.

The refinement connects to a live question. In the Alon–Tarsi approach [1] to Rota's basis conjecture, the Alon–Tarsi constant (the difference between the numbers of even and odd Latin squares of order n) is expressed as a signed sum over 0-1 matrices of $(-1)^{|A|}\det(A)^n$ [7], i.e. as a Fourier–Walsh coefficient of $\det^n$; since det expands over permutation matrices, the expansion is organized by how far a permutation matrix sits from the diagonal. In the Polymath 12 discussion of Rota's basis conjecture the imbalance (1) served as the computable model case of this family: in the comment thread of [14], G. Kalai (14 March 2017) proposes exactly this even/odd derangement imbalance as the heuristic model case, and T. Chow (28 March

2017) records the next cases as OEIS A098276 [12]; the heuristic was left unfinished there. Theorem 2 closes the model case completely: it is the exact graded refinement of (1) by distance from the diagonal. Conversely, reading (8) probabilistically yields an exact sign-randomization principle (Corollary 4) and a precise asymptotic independence statement between the sign and the excedance number of a random derangement (Corollary 5).

Section 2 states and proves the bivariate identity; the proof is a five-line evaluation of the determinant, in the same spirit as the determinant proof of (1) above. Section 3 derives the derangement fiber and its rigidity properties, draws two consequences — an exact sign-randomization principle and the $q=-1$ evaluation — and records the fact that the phenomenon is special to the excedance statistic: the corresponding inv- and maj-weighted signed enumerators are irregular already at $n=4$ (Table 2). Section 4 develops the type-B analogue: the standard sign character yields a vanishing full-group identity and the constant derangement fiber $(-1)^n$, while the twist by negation parity — identified, via an absolute-sign identity, with the sign of the underlying permutation — carries the true analogue $2(1-q)^{n-1}$ with a four-fold refinement by negation residues; both are proved by a two-line determinant evaluation, and on the half $\{\pi(1)<0\}$ the cancellation of both characters is proved by an explicit value-swap involution. Section 5 carries the graded programme out for the Alon–Tarsi constant itself: graded by sparsity, its layers vanish identically at odd orders and are divisible by $n!$ at even orders, the nonzero layers carry the sign $(-1)^k$, and the two lowest layers are computed in closed form. Section 6 poses the problems of a bijective proof of the rigidity statement, of a $q=-1$ involution, of a bijective proof of the twisted type-B identities, and of a structural description of the level algebra generated by the signed excedance layers; that algebra is shown there to contain the group algebra of the long cycle, to be semisimple, and to carry the augmentation and the sign among its characters, whose eigenvalues are the signed and the unsigned derangement enumerators, as well as a cyclotomic family of candidates $\chi_a(\varepsilon_{n,k})=\zeta_n^{ak}$, proved to be characters for $n=3,5$ and proved not to be for $n=4,6$; what remains open is its commutativity, the rest of its spectrum and its dimension. Appendix A describes the accompanying script with which all enumerative data in this note were verified.

2. The bivariate identity

Throughout, permutations are written in one-line notation, $\pi=(\pi(1),\dots,\pi(n))$, and

$$\mathrm{fix}(\pi)=\#\{i:\pi(i)=i\},\qquad \mathrm{exc}(\pi)=\#\{i:\pi(i)>i\}.$$

Theorem 1. *For every $n\ge 1$,*

$$\sum_{\pi\in S_n}\mathrm{sgn}(\pi)\,y^{\mathrm{fix}(\pi)}\,q^{\mathrm{exc}(\pi)}=\det T_n(y,q)=\frac{(y-q)^n-q(y-1)^n}{1-q}, \tag{5}$$

where $T_n(y,q)$ is the $n\times n$ matrix

$$T_n(y,q)=\begin{pmatrix} y & q & \cdots & q\\ 1 & y & \ddots & \vdots\\ \vdots & \ddots & \ddots & q\\ 1 & \cdots & 1 & y\end{pmatrix},\qquad t_{ij}=\begin{cases} y & j=i,\\ q & j>i,\\ 1 & j<i.\end{cases}$$

We write $D_n(y,q):=\det T_n(y,q)$ for either side of (5). The right-hand side has a removable singularity at $q=1$, where its limit equals $(y-1)^{n-1}(y+n-1)$. The identity also holds for $n=0$, both sides being 1.

Proof. By the Leibniz formula,

$$\det T_n(y,q) = \sum_{\pi\in S_n} \operatorname{sgn}(\pi) \prod_{i=1}^{n} t_{i,\pi(i)},$$

and the entry $t_{i,\pi(i)}$ equals y, q or 1 according to $\pi(i)=i$, $\pi(i)>i$, $\pi(i)<i$; hence the product equals $y^{\mathrm{fix}(\pi)}q^{\mathrm{exc}(\pi)}$, which proves the first equality in (5).

For the evaluation, write $D_n = D_n(y,q)$ and perform the column operations $C_j \leftarrow C_j - C_{j-1}$ for $j = n, n-1, \dots, 2$. Since $t_{i,j} - t_{i,j-1}$ equals $q-y$ in row $i=j-1$, equals $y-1$ in row $i=j$, and vanishes in every other row, the new column j $(2\le j\le n)$ is $(q-y)e_{j-1} + (y-1)e_j$, while $C_1 = (y,1,\dots,1)^T$ is unchanged. The last row of the resulting matrix has exactly two non-zero entries: 1 in column 1 and $y-1$ in column n. Expanding the determinant along this row,

$$D_n = (-1)^{n+1} M_{n,1} + (y-1) M_{n,n},$$

where $M_{n,n}$ is precisely the matrix obtained from $T_{n-1}(y,q)$ by the same operations, so $M_{n,n} = D_{n-1}$; and $M_{n,1}$ is lower bidiagonal of size $n-1$ with diagonal entries $q-y$ and subdiagonal entries $y-1$, so $M_{n,1} = (q-y)^{n-1}$. Since $(-1)^{n+1}(q-y)^{n-1} = (y-q)^{n-1}$,

$$D_n = (y-q)^{n-1} + (y-1)D_{n-1}, \qquad D_1 = y. \tag{6}$$

One checks that

$$D_n^* := \frac{(y-q)^n - q(y-1)^n}{1-q}$$

satisfies $D_1^* = y$ and $D_n^* - (y-1)D_{n-1}^* = (y-q)^{n-1}$: indeed the difference equals $\big[(y-q)^n - (y-1)(y-q)^{n-1}\big]/(1-q) = (y-q)^{n-1}\big[(y-q)-(y-1)\big]/(1-q) = (y-q)^{n-1}$. By (6), $D_n = D_n^*$ for all n.

Finally, the numerator $N(y,q) = (y-q)^n - q(y-1)^n$ of D_n^* vanishes at $q=1$, while $N'(y,1) = -n(y-1)^{n-1} - (y-1)^n$, so the limit of D_n^* at $q=1$ is $-N'(y,1) = (y-1)^{n-1}(y+n-1)$, which is also $\det T_n(y,1) = \det(J_n + (y-1)I_n)$. □

Remark 1. Four classical identities are specializations of (5).

(i) $q=1$ gives (2), the signed fixed-point enumerator (studied in [13]; at $y=0$ it is (1)).
(ii) $y=1$ gives (3), the signed Eulerian identity of Désarménien and Foata [6]. Note that $T_n(1,q)$ is the staircase matrix with q above the diagonal and 1 elsewhere, so Theorem 1 also provides a four-line determinant proof of (3).
(iii) $q=1$, $y=0$ gives $\det(J_n - I_n) = (-1)^{n-1}(n-1)$, the determinant proof of (1).
(iv) $y=q=t$ gives the *weak excedance* version: since $\#\{i : \pi(i)\ge i\} = \mathrm{exc}(\pi) + \mathrm{fix}(\pi)$,

$$\sum_{\pi\in S_n} \operatorname{sgn}(\pi)\, t^{\#\{i:\pi(i)\ge i\}} = D_n(t,t) = t(t-1)^{n-1},$$

once more binomial up to sign; see also [15, Thm. 6].

More generally, any statistic that compares $\pi(i)$ with i entry by entry — equivalently, any statistic of the form $\sum_i f(i,\pi(i))$ with $f(i,j)$ depending only on i and on the sign of $j-i$ — is obtained from a matrix of the same shape as $T_n(y,q)$, and the column operations of the proof of Theorem 1 evaluate it; specializing $y=0$ then gives the signed derangement enumerator of that statistic. The rigidity of Theorem 2 is thus a property of this whole class of statistics, not of exc alone; Section 2.1 carries out one further instance and Remark 5 records what happens outside the class.

2.1. **Infusing the fixed-point weight into the Sivasubramanian catalog.** The determinant method of Theorem 1 does not depend on the excedance statistic. Sivasubramanian [15] systematically computes signed enumerators over S_n and over $\mathcal{D}_n$ for a catalog of entry-wise statistics — exc, skew-exc, weak-exc, weak-skew-exc, the excedance sum $\mathrm{ESum}(\pi) = \sum_{i\in\mathrm{ExcSet}(\pi)} i$, the excedance length, and weighted versions — always with the diagonal weight pinned at 1, or removed entirely by zeroing it. Infusing the fixed-point weight y into the diagonal of any of his matrices and repeating the column operation of Theorem 1 yields a y-analogue of the corresponding entry of his catalog; the column operation survives because the statistic is entry-wise. We record one representative example and note that the other entries are similar.

Proposition 1. *For $n \geq 1$ let $\mathrm{ESum}(\pi) = \sum_{i:\pi(i)>i} i$ and*

$$\mathrm{ES}_n(y,q) := \sum_{\pi\in S_n} \mathrm{sgn}(\pi)\, y^{\mathrm{fix}(\pi)}\, q^{\mathrm{ESum}(\pi)} = \det P_n(y,q),$$

where P_n has (i,i)-entry y, (i,j)-entry q^i for $j>i$, and (i,j)-entry 1 for $j<i$. Then

$$\mathrm{ES}_n(y,q) = y(y-1)^{n-1} + \sum_{j=2}^{n}(y-1)^{n-j}\prod_{c=1}^{j-1}\bigl(y-q^c\bigr).$$

In particular $\mathrm{ES}_n(1,q) = \prod_{k=1}^{n-1}(1-q^k)$ and $\mathrm{ES}_n(0,q) = (-1)^{n-1}\sum_{j=1}^{n-1} q^{\binom{j+1}{2}}$, recovering [15, Thm. 8 and Thm. 19].

Proof. By the Leibniz formula $\mathrm{ES}_n = \det P_n$. Perform $C_j \leftarrow C_j - C_{j-1}$ for $j = n,\dots,2$: the new column j has entry $q^{j-1}-y$ in row $j-1$, entry $y-1$ in row j, and zeros elsewhere. The last row therefore has exactly two non-zero entries, 1 in column 1 and $y-1$ in column n, and expanding along it gives $\mathrm{ES}_n = (-1)^{n+1}M_{n,1} + (y-1)\mathrm{ES}_{n-1}$, where $M_{n,1}$ (the minor obtained by deleting row n and column 1) is lower bidiagonal of size $n-1$ with diagonal entries q^c-y, $1\leq c\leq n-1$, and subdiagonal entries $y-1$; hence $M_{n,1} = \prod_{c=1}^{n-1}(q^c-y)$. Since

$$(-1)^{n+1}\prod_{c=1}^{n-1}(q^c-y) = (-1)^{n+1}(-1)^{n-1}\prod_{c=1}^{n-1}(y-q^c) = \prod_{c=1}^{n-1}(y-q^c),$$

$$\mathrm{ES}_n = (y-1)\,\mathrm{ES}_{n-1} + \prod_{c=1}^{n-1}\bigl(y-q^c\bigr), \qquad \mathrm{ES}_1 = y, \tag{7}$$

and unrolling (7) gives the displayed sum. The specializations at $y=1$ and $y=0$ follow by substitution — at $y=1$ only the $j=n$ term survives, at $y=0$ the factor $\prod_{c=1}^{j-1}(-q^c)$ contributes $(-1)^{j-1}q^{\binom{j}{2}}$ — and match [15, Thm. 8, Thm. 19]. □

Corollary 1 (weighted version)**.** *More generally, for positive integers $w_1,\dots,w_{n-1}$ (Sivasubramanian's weighted excedance-sum statistic: an excedance at position i carries weight q^{w_i}), the y-infused enumerator satisfies*

$$\sum_{\pi\in S_n}\mathrm{sgn}(\pi)\, y^{\mathrm{fix}(\pi)} q^{\sum_{i:\pi(i)>i} w_i} = y(y-1)^{n-1} + \sum_{j=2}^{n}(y-1)^{n-j}\prod_{c=1}^{j-1}\bigl(y-q^{w_c}\bigr),$$

recovering [15, Thm. 20] *at $y=0$.*

Proof. Identical: the column operation now produces the diagonal entries $q^{w_c}-y$, $1\leq c\leq n-1$, in the minor $M_{n,1}$ and the entries $y-1$ on its subdiagonal, and the recursion unrolls the same way. □

Proposition 1 and Corollary 1 also delimit where the method stops working. The difference $C_j - C_{j-1}$ is two-sparse precisely when the above-diagonal weights are *constant along rows*, that is, when the statistic has the form $\mathrm{st}(\pi) = \sum_{i:\pi(i)>i} f(i)$ for a weight function f; every such f is then covered by Corollary 1 with $w_c = f(c)$, and each choice produces a family interpolating between two of Sivasubramanian's theorems (the specializations $y = 1$ and $y = 0$). A statistic that depends on the *value* $\pi(i)$ — for instance the excedance length $\mathrm{ELen}(\pi) = \sum_{i:\pi(i)>i}\big(\pi(i) - i\big)$, whose natural matrix has (i, j)-entry q^{j-i} above the diagonal — gives $t_{i,j} - t_{i,j-1} = q^{j-i} - q^{j-1-i} \neq 0$ in every row $i < j-1$, so the column operation does not reduce the matrix and the computation fails. In this sense the fixed-point weight can be infused uniformly into the row-constant part of the catalog of [15], but not into all of it; we have not found a matrix model for the value-dependent statistics that admits the weight y.

3. The derangement fiber

Theorem 2 (Mantaci–Rakotondrajao [9]; cf. [15])**.** *For every $n \geq 1$,*

$$F_n(q) := \sum_{\pi\in\mathcal{D}_n} \mathrm{sgn}(\pi)\, q^{\mathrm{exc}(\pi)} = (-1)^{n-1}\big(q + q^2 + \cdots + q^{n-1}\big) = (-1)^{n-1}\frac{q(1-q^{n-1})}{1-q}. \tag{8}$$

(For $n = 1$ the sum is empty and $F_1 = 0$; for $n = 0$ we set $F_0 = 1$, and (8) holds for $n = 0$ as well with the convention $q^0 = 1$.)

Proof. Set $y = 0$ in Theorem 1. Since $0^{\mathrm{fix}(\pi)}$ vanishes unless $\mathrm{fix}(\pi) = 0$, the left-hand side becomes $F_n(q)$, while the right-hand side becomes

$$\frac{(0-q)^n - q(0-1)^n}{1-q} = \frac{(-1)^n q^n - q(-1)^n}{1-q} = (-1)^n\frac{q(q^{n-1}-1)}{1-q} = (-1)^{n-1}\frac{q(1-q^{n-1})}{1-q}. \qquad \square$$

Corollary 2. *For all $n \geq 2$ and every $1 \leq k \leq n-1$,*

$$E_{n,k} - O_{n,k} = (-1)^{n-1},$$

where $E_{n,k}$ and $O_{n,k}$ count the even and odd derangements of $[n]$ with $\mathrm{exc}(\pi) = k$. Equivalently, $E_{n,k} = \big(d_{n,k} + (-1)^{n-1}\big)/2$, where

$$d_{n,k} := \#\{\pi \in \mathcal{D}_n : \mathrm{exc}(\pi) = k\}$$

is the excedance distribution on derangements, the coefficient of q^k in $d_n(q) := \sum_{\pi\in\mathcal{D}_n} q^{\mathrm{exc}(\pi)}$.

Proof. $F_n(q) = \sum_k (E_{n,k} - O_{n,k})q^k$; compare coefficients with (8). The expression for $E_{n,k}$ follows from $E_{n,k} + O_{n,k} = d_{n,k}$. $\square$

Corollary 3. *For $q = 1$, Theorem 2 recovers the classical identity $E_n - O_n = (-1)^{n-1}(n-1)$.*

Remark 2 (an independent route). Theorem 2 can also be derived without the determinant, from (3) alone. The key standard fact is that deleting all fixed points of $\pi \in S_n$ and standardizing preserves both $\mathrm{sgn}(\pi)$ and $\mathrm{exc}(\pi)$: the sign is unchanged because π is the extension of its restriction by fixed points, and the excedance count is unchanged because the standardization is order-preserving on $[n] \setminus \mathrm{fix}(\pi)$. Every $\pi \in S_n$ whose fixed-point set has size $n - m$ thus contributes, in (3), a term $F_m(q)$, and the fixed set may be chosen in $\binom{n}{m}$ ways; hence, with $F_0 = 1$,

$$G_n := \sum_{m=0}^{n}\binom{n}{m}F_m(q) = \begin{cases} 1 & n = 0, \\ (1-q)^{n-1} & n \geq 1. \end{cases} \tag{9}$$

Binomial inversion reads $F_n = \sum_{m=0}^{n} \binom{n}{m}(-1)^{n-m}G_m$; the term $m = 0$ must be kept separate, since $G_0 = 1$ and not $(1-q)^{-1}$, and gives

$$F_n = (-1)^n + \frac{1}{1-q}\sum_{m=1}^{n}\binom{n}{m}(-1)^{n-m}(1-q)^m = (-1)^n + \frac{(-q)^n - (-1)^n}{1-q},$$

which is (8): clearing the denominator, $(1-q)F_n = (-1)^n(1-q)+(-q)^n-(-1)^n = (-1)^n\big(q^n-q\big) = (-1)^{n-1}q(1-q^{n-1})$. The same inversion, applied with the factor 2 of Theorem 3, is the engine of Theorem 4.

Corollary 4 (sign-randomization)**.** *Let $n \geq 2$, $1 \leq k \leq n-1$, and let π be a uniformly random element of $\{\pi \in \mathcal{D}_n : \mathrm{exc}(\pi) = k\}$. Then*

$$\Pr[\pi \text{ even}] \;=\; \frac{1}{2} + \frac{(-1)^{n-1}}{2\,d_{n,k}}, \qquad d_{n,k} := \#\{\pi \in \mathcal{D}_n : \mathrm{exc}(\pi) = k\}.$$

Proof. $\Pr[\pi \text{ even}] = E_{n,k}/(E_{n,k}+O_{n,k})$; substitute Corollary 2 and $E_{n,k}+O_{n,k} = d_{n,k}$. □

Remark 3 (shape inheritance). The polynomial $d_n(q) = \sum_k d_{n,k}q^k$ has symmetric coefficients, $d_{n,k} = d_{n,n-k}$ (visible already in $d_{4,1} = d_{4,3} = 1$, $d_{4,2} = 7$). Indeed the reverse-complement $\pi \mapsto \bar{\pi} := w_0\pi w_0$ of Proposition 2 below permutes $\mathcal{D}_n$ — it preserves fixed points, since $\bar{\pi}(i) = i$ is equivalent to $\pi(n+1-i) = n+1-i$ — and sends exc to $n-$exc. Hence the two sequences $(E_{n,k})$ and $(O_{n,k})$ are symmetric in k as well, and (8) exhibits the excedance distribution as a sum $d_{n,k} = E_{n,k} + O_{n,k}$ of two symmetric twin sequences differing by the constant $(-1)^{n-1}$. Any shape information available for $d_{n,k}$ — unimodality-type properties, or limit theorems such as those established for the maj-weighted q-derangement numbers [5] — therefore transfers to the pair $(E_{n,k}, O_{n,k})$ up to the bounded shift $(-1)^{n-1}/2$.

Corollary 5 (asymptotic independence of sign and exc)**.** *Let $k(n)$ be any sequence with $d_{n,k(n)} \to \infty$, and let π be a uniformly random derangement of $[n]$ conditioned on $\mathrm{exc}(\pi) = k(n)$. Then*

$$\Pr[\pi \text{ even}] = \frac{1}{2} + \frac{(-1)^{n-1}}{2\,d_{n,k(n)}} \longrightarrow \frac{1}{2},$$

with signed error exactly $\pm\,(2d_{n,k(n)})^{-1}$ at every n. The coefficients $d_{n,k}$ are of order $n!/(e\sqrt{n})$ over the central $\sqrt{n}$-window of the excedance distribution, so every sequence $k(n)$ in that window has $d_{n,k(n)} \to \infty$ and the display applies to it; for the maj*-weighted q-derangement numbers this order of magnitude follows from the limit theorem of* [5]*, and the transfer from* maj *to* exc *on $\mathcal{D}_n$ is the expected analogue.*

Corollary 6 (joint sign-randomization)**.** *For $0 \leq f \leq n$ and every $e \geq 0$, let $c_{n,f,e}$ denote the signed count*

$$\#\{\pi \in S_n : \mathrm{fix}(\pi) = f,\ \mathrm{exc}(\pi) = e,\ \pi \text{ even}\} - \#\{\pi \in S_n : \mathrm{fix}(\pi) = f,\ \mathrm{exc}(\pi) = e,\ \pi \text{ odd}\}.$$

Then $c_{n,f,e} = \binom{n}{f}(-1)^{n-f-1}$ if $1 \leq e \leq n-f-1$, equals 1 for $(f,e) = (n,0)$, and vanishes otherwise. Equivalently: conditional on $\mathrm{fix}(\pi) = f$ and $\mathrm{exc}(\pi) = e$ (with nonempty conditioning class), the sign of a uniformly random permutation of $[n]$ deviates from fair by exactly $\binom{n}{f}^{-1}(-1)^{n-f-1}$ in signed probability.

Proof. By the decomposition of Remark 2 (delete the f fixed points; both sgn and exc survive), $c_{n,f,e} = \binom{n}{f}\,[q^e]\,F_{n-f}(q)$; substitute Theorem 2. The case $(f,e) = (n,0)$ is the identity permutation. □

Remark 4 (unconditional and conditional bias). Corollaries 4, 5 and 6 are three readings of one phenomenon, at increasing levels of conditioning. Unconditionally, $\Pr[\pi \in \mathcal{D}_n \text{ even}] = \frac{1}{2} + \frac{(-1)^{n-1}(n-1)}{2\,!n}$ (here $!n$ is the derangement number, $!n = n!\sum_{j\le n}(-1)^j/j!$): the bias is exponentially small. Conditioning on $\mathrm{exc} = k$ alone, the bias is exactly $\pm 1/(2d_{n,k})$ (Corollary 4), which tends to zero in the central $\sqrt{n}$-window but is $O(1)$ at the extreme levels, where $d_{n,k} \in \{1, 2\}$; conditioning on the pair $(\mathrm{fix}, \mathrm{exc})$, the bias is $\binom{n}{f}^{-1}(-1)^{n-f-1}$ (Corollary 6). The large and moderate deviation regimes of $d_{n,k}$ away from the center should be governed by the asymptotics of the q-derangement coefficients, known for the maj-weighting [5].

Corollary 7 ($q = -1$)**.** *For every $n \ge 1$,*

$$F_n(-1) = \sum_{\pi\in\mathcal{D}_n} (-1)^{\mathrm{inv}(\pi)+\mathrm{exc}(\pi)} = \begin{cases} 1 & n \text{ even}, \\ 0 & n \text{ odd}. \end{cases}$$

Proof. Evaluate (8) at $q = -1$; the alternating sum $\sum_{k=1}^{n-1}(-1)^k$ equals 0 when $n-1$ is even and -1 when $n-1$ is odd, and $(-1)^{n-1}(-1) = (-1)^n$. □

The odd case admits a bijective explanation.

Proposition 2 (the odd case of Corollary 7, bijectively)**.** *Let n be odd and let $w_0 \in S_n$ be the reversal $i \mapsto n+1-i$. The reverse-complement map $\pi \mapsto \bar\pi := w_0\pi w_0$ is a fixed-point-free involution of $\mathcal{D}_n$, and for every $\pi \in \mathcal{D}_n$,*

$$\mathrm{inv}(\bar\pi) = \mathrm{inv}(\pi), \qquad \mathrm{exc}(\bar\pi) = n - \mathrm{exc}(\pi), \qquad (-1)^{\mathrm{inv}(\bar\pi)+\mathrm{exc}(\bar\pi)} = -(-1)^{\mathrm{inv}(\pi)+\mathrm{exc}(\pi)}.$$

Hence for odd n the vanishing in Corollary 7 is witnessed by a perfect pairing.

Proof. $\bar\pi(i) = i$ would force $\pi(n+1-i) = n+1-i$, a fixed point of π; since $\pi \in \mathcal{D}_n$ has none, $\bar\pi \in \mathcal{D}_n$. The map is visibly an involution. For the statistics: each inversion pair (i, j), $i < j$, $\pi(i) > \pi(j)$ of π corresponds to the pair $(n+1-j, n+1-i)$, $n+1-j < n+1-i$, $n+1-\pi(j) > n+1-\pi(i)$ of $\bar\pi$, so $\mathrm{inv}(\bar\pi) = \mathrm{inv}(\pi)$; and using $\pi(j) \ne j$,

$$\mathrm{exc}(\bar\pi) = \#\{i : n+1-\pi(n+1-i) > i\} = \#\{j : \pi(j) < j\} = n - \mathrm{exc}(\pi).$$

Finally, if $\bar\pi = \pi$ then at the middle position $m = (n+1)/2$ we get $\pi(m) = m$, again impossible; so the involution is fixed-point-free, and $(-1)^{\mathrm{inv}(\bar\pi)+\mathrm{exc}(\bar\pi)} = (-1)^{\mathrm{inv}(\pi)+n-\mathrm{exc}(\pi)} = -(-1)^{\mathrm{inv}(\pi)+\mathrm{exc}(\pi)}$ because n is odd. □

For even n the reverse-complement *preserves* the weight $(-1)^{\mathrm{inv}\,+\,\mathrm{exc}}$, so a different mechanism is needed. It turns out that the split suggested by computation is a theorem. Write $\bar\pi = w_0\pi w_0$ and recall the identification of reverse-complementary permutations with B_m (for $n = 2m$): π acts on the m pairs $\{i, 2m+1-i\}$ as a signed permutation $\sigma \in B_m$, and π is a derangement of $[2m]$ exactly when σ has no absolute fixed point ($\sigma(a) \ne a$ for all a; a diagonal pair with negative orientation is allowed and is fixed-point-free). Under this identification the reverse-complementary derangements are counted by the type-B derangement numbers $1, 5, 29, 233, \ldots$ [11].

Lemma 1 (excedances of reverse-complementary derangements)**.** *Every reverse-complementary derangement of $[2m]$ has exactly m excedances.*

Proof. For π reverse-complementary, $\pi(a) + \pi(2m+1-a) = 2m+1 = a + (2m+1-a)$, so

$$[\pi(a) > a] + [\pi(2m+1-a) > 2m+1-a] = 1$$

unless $\pi(a) = a$, which cannot occur in a derangement. Summing over the m pairs gives $\mathrm{exc}(\pi) = m$. □

Lemma 2 (the ambient sign)**.** *Let $\pi \in B_m \subset S_{2m}$ (reverse-complementary) with signed one-line $\sigma = (\sigma(1), \dots, \sigma(m))$. Then $\operatorname{sgn}_{S_{2m}}(\pi) = (-1)^{\#\{a:\sigma(a)<0\}}$.*

Proof. Both sides are characters of B_m, and B_m is generated by the pair negations $a \mapsto -a$ and the pair transpositions. A pair negation lifts to a single transposition in S_{2m} (sign -1) and changes $\#\{a : \sigma(a) < 0\}$ by one; a pair transposition lifts to a product of two transpositions (sign $+1$) and leaves the negation count unchanged. □

Proposition 3 (the even case, structurally)**.** *Let $n = 2m$, and split $\mathcal{D}_{2m}$ into the reverse-complementary derangements ($\bar{\pi} = \pi$) and the remaining ones. Then*

$$\sum_{\substack{\pi \in \mathcal{D}_{2m} \\ \bar{\pi} = \pi}} (-1)^{\operatorname{inv}(\pi)+\operatorname{exc}(\pi)} = +1 \qquad \text{and} \qquad \sum_{\substack{\pi \in \mathcal{D}_{2m} \\ \bar{\pi} \neq \pi}} (-1)^{\operatorname{inv}(\pi)+\operatorname{exc}(\pi)} = 0. \tag{10}$$

Consequently the reverse-complementary layer alone accounts for the entire value $F_{2m}(-1) = 1$ of Corollary 7.

Proof. Let $\chi_{\mathrm{neg}}(\sigma) = (-1)^{\#\{a:\sigma(a)<0\}}$ denote the negation-parity character of B_m; by Lemma 2 the ambient sign of a reverse-complementary permutation is χ_{neg} of its signed one-line. By Lemma 1, for $\pi \in \mathcal{D}_{2m}$ reverse-complementary,

$$(-1)^{\operatorname{inv}(\pi)+\operatorname{exc}(\pi)} = (-1)^m \cdot \operatorname{sgn}_{S_{2m}}(\pi) = (-1)^m \, \chi_{\mathrm{neg}}(\sigma).$$

Now sum over $\mathcal{D}(B_m)$ by inclusion–exclusion over absolute fixed points:

$$\sum_{\sigma \in \mathcal{D}(B_m)} \chi_{\mathrm{neg}}(\sigma) = \sum_{S \subseteq [m]} (-1)^{|S|} \sum_{\substack{\sigma \in B_m \\ \sigma(a) = a \ (a \in S)}} \chi_{\mathrm{neg}}(\sigma).$$

The permutations fixing S pointwise form a subgroup isomorphic to $B_{m-|S|}$, on which χ_{neg} restricts to the negation-parity character; this character is nontrivial as long as $m - |S| \geq 1$, so every inner sum vanishes except for $S = [m]$, where it equals $\chi_{\mathrm{neg}}(\mathrm{id}) = 1$. Hence $\sum_{\mathcal{D}(B_m)} \chi_{\mathrm{neg}} = (-1)^m$, and the reverse-complementary layer contributes $(-1)^m \cdot (-1)^m = +1$. The non-reverse-complementary layer equals $F_{2m}(-1) - 1 = 0$ by Corollary 7 (which follows from Theorem 1). □

The two layers leave one combinatorial desideratum open (Problem 3): an explicit weight-reversing involution on the non-reverse-complementary derangements, explaining their cancellation directly rather than by subtraction.

n	$F_n(q)$
2	$-q$
3	$q + q^2$
4	$-q - q^2 - q^3$
5	$q + q^2 + q^3 + q^4$
6	$-(q + q^2 + q^3 + q^4 + q^5)$
7	$q + q^2 + q^3 + q^4 + q^5 + q^6$

TABLE 1. The signed excedance enumerator of $\mathcal{D}_n$ for small n.

Remark 5 (the phenomenon is special to exc). The proof of Theorem 1 uses that $\operatorname{exc}(\pi)$ is an *entry-wise* statistic, $\operatorname{exc}(\pi) = \sum_i [\pi(i) > i]$, so that the weights $q^{\operatorname{exc}(\pi)}$ factor over the entries selected by π. The statistics inv and maj are not entry-wise, and their signed derangement

enumerators are not rigid; see Table 2. This is in marked contrast with the *unsigned* maj-weighted derangement polynomials of Wachs [19], whose coefficients are nonnegative, symmetric and unimodal. (Curiously the signed maj-enumerator happens to coincide with $F_n(q)$ at $n = 5$; the coincidence is accidental, as the cases $n = 4$ and $n = 6$ already show.)

n	$\sum_{\pi\in\mathcal{D}_n} \operatorname{sgn}(\pi)\, q^{\mathrm{inv}(\pi)}$	$\sum_{\pi\in\mathcal{D}_n} \operatorname{sgn}(\pi)\, q^{\mathrm{maj}(\pi)}$
4	$q^2 - 4q^3 + q^4 - 2q^5 + q^6$	$-q - 2q^3 - q^5 + q^6$
5	$-4q^3 + 8q^4 - 4q^5 + 10q^6 - 10q^7 + 6q^8 - 2q^9$	$q + q^2 + q^3 + q^4$
6	$-q^3 + 12q^4 - 18q^5 + 16q^6 - 35q^7 + 44q^8$	$-q - 3q^3 - 4q^5 + 2q^6 - 3q^7 + 4q^8$
	$-47q^9 + 40q^{10} - 25q^{11} + 14q^{12} - 8q^{13} + 4q^{14} - q^{15}$	$-q^9 + 4q^{10} - 2q^{11} + 2q^{12} - 2q^{13} - q^{15}$

TABLE 2. The inv- and maj-weighted signed derangement enumerators are not rigid: coefficients other than ± 1 appear already at $n = 4$.

Remark 6 (three statistics for the price of two). Writing $\mathrm{anti}(\pi) = \#\{i : \pi(i) < i\}$ for the number of anti-excedances, every permutation satisfies $\mathrm{exc}(\pi) + \mathrm{anti}(\pi) + \mathrm{fix}(\pi) = n$. Hence (5) determines the joint signed distribution of all three statistics: the triple enumerator is

$$\sum_{\pi\in S_n} \operatorname{sgn}(\pi)\, y^{\mathrm{fix}(\pi)} q^{\mathrm{exc}(\pi)} z^{\mathrm{anti}(\pi)} = z^n\, D_n\Big(\frac{y}{z}, \frac{q}{z}\Big),$$

where $D_n(y, q)$ is the polynomial of Theorem 1. In this sense the identity of Theorem 1 is complete.

4. The type-B analogue

Let B_n be the hyperoctahedral group: bijections π of $\{\pm 1, \dots, \pm n\}$ with $\pi(-i) = -\pi(i)$, written in one-line notation as $(\pi(1), \dots, \pi(n))$ with $\pi(i) \in \{\pm 1, \dots, \pm n\}$. The *type-B length* is

$$l_B(\pi) = \#\{i < j : \pi(i) > \pi(j)\} + \#\{i \le j : \pi(i) + \pi(j) < 0\}, \qquad \operatorname{sgn}_B(\pi) = (-1)^{l_B(\pi)},$$

see [4, Prop. 8.1.1]; thus sgn_B is the sign character of B_n, equivalently the determinant of the signed permutation matrix of π, and we warn the reader that it is *not* the parity of the linear extension to S_{2n} (the two differ already at $(2, 1) \in B_2$). Alongside it we use the *negation-parity character* $\chi_{\mathrm{neg}}(\pi) = (-1)^{\#\{i:\pi(i)<0\}}$, already met in Lemma 2, and their product

$$\operatorname{sgn}_B^*(\pi) := \operatorname{sgn}_B(\pi)\, \chi_{\mathrm{neg}}(\pi) = (-1)^{l_B(\pi)+\#\{i:\pi(i)<0\}},$$

the *twist* of sgn_B, which we also write $\tau(\pi)$. These are three of the exactly four one-dimensional characters of B_n (the fourth is trivial), and the type-B story told below is genuinely character-sensitive. The *absolute excedance number* is $\mathrm{exc}_B(\pi) = \#\{i : \pi(i) > i\}$, and the *absolute derangements* are $\mathcal{D}(B_n) = \{\pi \in B_n : \pi(i) \ne i \text{ for all } i\}$ (no condition is imposed on $\pi(i) = -i$).

Convention. In this section permutations are composed *left to right*, $(\sigma\rho)(i) = \rho(\sigma(i))$, so that a right multiplication $\sigma \mapsto \sigma\rho$ acts on the *values* of σ and leaves its positions alone. This is the convention under which the involutions of Propositions 4 and 5 preserve exc_B; with the opposite convention they would not. Since sgn_B, χ_{neg} and τ are homomorphisms, they remain multiplicative for either convention.

Which character continues the type-A story? The naive candidate is not viable:

Proposition 4 (the standard-character vanishing)**.** *For every $n \ge 1$,*

$$\sum_{\pi\in B_n} \operatorname{sgn}_B(\pi)\, q^{\mathrm{exc}_B(\pi)} = 0.$$

Proof. Let $g = (-1, 2, \dots, n) \in B_n$, the simple reflection s_0 that negates the value 1, so that $\operatorname{sgn}_B(g) = -1$ and $g^2 = e$. By the convention above, σg is obtained from σ by applying g to its values, i.e. by interchanging the entries 1 and -1 wherever they occur; hence $\sigma \mapsto \sigma g$ is an involution of B_n, and it is fixed-point-free because $\sigma g = \sigma$ would force $g(\sigma(i)) = \sigma(i)$ for every i, hence $g = e$ since σ is onto. It preserves exc_B: the only values touched are ± 1, and neither 1 nor -1 exceeds any position $i \geq 1$. Since sgn_B is multiplicative, $\operatorname{sgn}_B(\sigma g) = -\operatorname{sgn}_B(\sigma)$, and each g-orbit pairs equal-excedance permutations of opposite sign. □

The twist itself admits a complete description, and it is not really a type-B object at all: it is the sign of the *absolute* permutation.

Lemma 3 (absolute-sign identity)**.** *Write* $|\pi| = (|\pi(1)|, \dots, |\pi(n)|) \in S_n$ *and let* inv *count inversions in the integer order. Then*

$$\operatorname{inv}(\pi) + \#\{i < j : \pi(i) + \pi(j) < 0\} \equiv \operatorname{inv}(|\pi|) \pmod 2,$$

and consequently

$$\tau(\pi) = \operatorname{sgn}(|\pi|).$$

Proof. By definition $l_B(\pi) = \operatorname{inv}(\pi) + \#\{i \leq j : \pi(i) + \pi(j) < 0\}$, and the diagonal terms $i = j$ of the second sum are exactly the negative values, whence $\tau(\pi) = (-1)^{l_B(\pi)+\operatorname{neg}(\pi)} = (-1)^{\operatorname{inv}(\pi)+\#\{i<j:\pi(i)+\pi(j)<0\}}$. For a pair $i < j$ put $(u, v) = (\pi(i), \pi(j))$ and $a = |u| \neq b = |v|$; the pair's contribution $[u > v] + [u + v < 0]$ is odd exactly when $a > b$: it equals $[a > b]$, $1 + [a < b]$, $[a > b]$, and $[a < b]+1$ in the four sign cases $(+,+)$, $(+,-)$, $(-,+)$, $(-,-)$, respectively. Summing over pairs, $\tau(\pi) = (-1)^{\#\{i<j:|\pi(i)|>|\pi(j)|\}} = \operatorname{sgn}(|\pi|)$. □

Theorem 3 (the twisted type-B identities)**.** *Let* $\operatorname{neg}(\pi) = \#\{i : \pi(i) < 0\}$. *For every* $n \geq 1$*:*

(i) $\displaystyle\sum_{\pi \in B_n} \tau(\pi)\, q^{\operatorname{exc}_B(\pi)} \;=\; 2(1-q)^{n-1}$;

(ii) *for every* $r \in \{0, 1, 2, 3\}$,

$$\sum_{\substack{\pi \in B_n \\ \operatorname{neg}(\pi) \equiv r \pmod 4}} \tau(\pi)\, q^{\operatorname{exc}_B(\pi)} = \begin{cases} (1-q)^{n-1} & r \equiv 0, 1 \pmod 4, \\ 0 & r \equiv 2, 3 \pmod 4. \end{cases}$$

Proof. Write $\pi(i) = \varepsilon_i\, w(i)$ with $w = |\pi| \in S_n$ and $\varepsilon_i \in \{\pm 1\}$; this parametrizes B_n by $S_n \times \{\pm 1\}^n$. By Lemma 3, $\tau(\pi) = \operatorname{sgn}(w)$; $\operatorname{exc}_B(\pi) = \sum_i [\varepsilon_i w(i) > i]$ is entry-wise; and $\operatorname{neg}(\pi) = \sum_i [\varepsilon_i = -]$. Since $-w(i) > i$ never holds, position i contributes $\big(q^{[w(i)>i]},\, 0\big)$ for $\varepsilon_i = +$ and $(1,\, 1)$ for $\varepsilon_i = -$. Summing over ε at fixed w multiplies by $\prod_i \big(q^{[w(i)>i]} + x\big)$, and summing over w with the Leibniz sign gives the two-variable refinement

$$\Phi_n(x, q) := \sum_{\pi \in B_n} \tau(\pi)\, x^{\operatorname{neg}(\pi)}\, q^{\operatorname{exc}_B(\pi)} = \det\big(X_n(x, q)\big), \qquad \big(X_n\big)_{i,j} = x + q^{[j>i]},$$

so X_n carries the constant entry x on and below the diagonal and $x + q$ above. (The entry-wise factorization of exc_B is the same phenomenon that drives Theorem 1; the point of Lemma 3 is that the twist joins it.) The row operations $R_i \leftarrow R_i - R_{i-1}$ for $i = n, \dots, 2$ produce the upper triangular matrix with diagonal $(x + 1,\, 1 - q, \dots, 1 - q)$ and first row $(x + 1,\, x + q, \dots, x + q)$, whence

$$\Phi_n(x, q) = (x + 1)(1 - q)^{n-1}.$$

Part (i) is $\Phi_n(1, q)$. For (ii), $[x^k]\Phi_n = (1-q)^{n-1}$ for $k = 0, 1$ and $[x^k]\Phi_n = 0$ for $k \geq 2$, and the class sum for residue r is $\sum_{k \equiv r \pmod 4} [x^k]\Phi_n$. □

$$\sum_{\pi\in B_n} \tau(\pi)\, q^{\mathrm{exc}_B(\pi)} = 2(1-q)^{n-1} \tag{11}$$

is the type-B analogue of the Désarménien–Foata identity (3), with the factor 2 in place of the type-A value. The full-group values for $n = 1, \dots, 5$ are 2; $2-2q$; $2-4q+2q^2$; $2-6q+6q^2-2q^3$; $2-8q+12q^2-8q^3+2q^4$ (at $n=7$ the level coefficients are $2, -12, 30, -40, 30, -12, 2 = 2\binom{6}{k}(-1)^k$). The same determinant settles the standard-character companion, by the substitution $x \mapsto -x$ in Φ_n:

$$\sum_{\pi\in B_n} \mathrm{sgn}_B(\pi)\, x^{\mathrm{neg}(\pi)}\, q^{\mathrm{exc}_B(\pi)} = (1-x)(1-q)^{n-1}, \tag{12}$$

so the negation-even and negation-odd classes carry sgn_B-weights $(1-q)^{n-1}$ and $-(1-q)^{n-1}$ respectively; at $x=1$ this recovers Proposition 4 algebraically.

Remark 7 (relation to colored excedance enumeration). Excedance statistics of colored permutation groups were studied by Bagno–Garber [2], whose signed enumerators weight by the cycle-count parity $(-1)^{\mathrm{cyc}}$ (the specialization $s=-1$ in their $P_{G_{r,n}}(q,t,s)$) and whose excedance statistic is augmented with the color sum, $\mathrm{exc}^{\mathrm{Abs}} = \mathrm{exc}(|\sigma|) + \mathrm{csum}(\sigma)$; for instance their type-D identity reads $P^{\mathrm{Clr}}_{D_n}(q,1,-1) = (1-q^2)^{n-1}$. Theorem 3 instead uses the twist τ and the plain absolute excedance number, with no color structure, and is therefore not a specialization of theirs.

The engine of Section 3 transfers verbatim. The deletion lemma holds in type B, for all three characters at once:

Lemma 4 (type-B deletion)**.** *Suppose $\pi \in B_n$ fixes i absolutely ($\pi(i)=i$), and let $\sigma \in B_{n-1}$ be the restriction of π to $\{\pm1,\dots,\pm n\}\setminus\{\pm i\}$, standardized. Then $\mathrm{sgn}_B(\pi) = \mathrm{sgn}_B(\sigma)$, $\chi_{\mathrm{neg}}(\pi) = \chi_{\mathrm{neg}}(\sigma)$, and $\mathrm{exc}_B(\pi) = \mathrm{exc}_B(\sigma)$; in particular the same holds for the twist sgn^*_B.*

Proof. Since π is a bijection and $\pi(i)=i$, π maps $\{\pm1,\dots,\pm n\}\setminus\{\pm i\}$ into itself, so the restriction is well defined; the standardized restriction is σ. For the sign, use the matrix realization of sgn_B as a determinant: the i-th row of the signed permutation matrix P_π is e_i, so expanding $\det P_\pi$ along this row gives $\det P_\pi = \det M$, where M is the minor obtained by deleting row and column i. The rows and columns of M are indexed by $\{\pm1,\dots,\pm n\}\setminus\{\pm i\}$ in their natural order, and the order-preserving relabeling onto $\{\pm1,\dots,\pm(n-1)\}$ is a simultaneous permutation of rows and columns, which leaves determinants invariant; the resulting matrix is P_σ. Hence $\mathrm{sgn}_B(\pi) = \mathrm{sgn}_B(\sigma)$. The deleted value $\pi(i)=i$ is positive, so the negation counts of π and σ agree and $\chi_{\mathrm{neg}}(\pi) = \chi_{\mathrm{neg}}(\sigma)$. For the excedances: the fixed point contributes 0 to $\mathrm{exc}_B(\pi)$, and the standardization ϕ is order-preserving, so on the remaining points $\pi(j) > j \iff \phi(\pi(j)) > \phi(j)$; the standardized restriction is precisely σ, giving $\mathrm{exc}_B(\pi) = \mathrm{exc}_B(\sigma)$. □

Consequently the twisted fiber formula follows unconditionally, by binomial inversion:

Theorem 4. *For every $n \geq 1$,*

$$N_n(q) := \sum_{\pi\in\mathcal{D}(B_n)} \mathrm{sgn}^*_B(\pi)\, q^{\mathrm{exc}_B(\pi)} = (-1)^{n-1}\big(1 + 2q + 2q^2 + \cdots + 2q^{n-1}\big). \tag{13}$$

In particular $N_n(1) = (-1)^{n-1}(2n-1)$: the numbers of even and odd absolute derangements in B_n differ by $2n-1$; and at coefficient level the imbalance is $(-1)^{n-1}$ at excedance level 0 and $2(-1)^{n-1}$ at every level $1 \leq k \leq n-1$.

Proof. By Lemma 4, every $\pi \in B_n$ whose absolute fixed-point set has size $n-m$ contributes, to the left-hand side of (11), the weight $N_m(q)$, and the fixed set may be chosen in $\binom{n}{m}$ ways; hence with $N_0 = 1$,

$$\sum_{m=0}^{n} \binom{n}{m} N_m(q) = 2(1-q)^{n-1}.$$

Binomial inversion, exactly as in (9) of Remark 2 ($N_n = \sum_m \binom{n}{m}(-1)^{n-m} G_m$ with $G_m := \sum_j \binom{m}{j} N_j$), gives, with $G_0 = N_0 = 1$ and $G_m = 2(1-q)^{m-1}$ for $m \geq 1$ — the $m = 0$ term contributing $(-1)^n$ —

$$\begin{aligned} N_n &= (-1)^n + \frac{2}{1-q} \sum_{m=1}^{n} \binom{n}{m} (-1)^{n-m} (1-q)^m \\ &= (-1)^n + \frac{2}{1-q}\big((-q)^n - (-1)^n\big) = (-1)^n \, \frac{2q^n - 1 - q}{1-q}, \end{aligned}$$

and $(1-q)(1 + 2q + \cdots + 2q^{n-1}) = 1 + q - 2q^n$ shows that this equals (13). The coefficient-level statement is read off from (13). □

Note the agreeable bookkeeping: type A has $n-1$ excedance levels, balanced up to exactly one derangement each; the twisted type-B fiber has n levels, balanced up to one derangement at level 0 (where the derangement $(-1, -2, \ldots, -n)$ lives, having no excedances) and up to a pair of derangements at every positive level.

With the standard character, by contrast, the fiber collapses to a constant:

Theorem 5 (the standard-character derangement fiber)**.** *For every $n \geq 1$,*

$$M_n(q) := \sum_{\pi \in \mathcal{D}(B_n)} \operatorname{sgn}_B(\pi)\, q^{\operatorname{exc}_B(\pi)} = (-1)^n, \tag{14}$$

a constant: at every excedance level $k \geq 1$ the even and odd absolute derangements of B_n cancel perfectly, and the entire imbalance $(-1)^n$ sits at level 0. It is the standard character, not the twist, that is rigid here.

Proof. Let $A_n(q) := \sum_{\pi \in B_n} \operatorname{sgn}_B(\pi)\, q^{\operatorname{exc}_B(\pi)}$, so that $A_n = 0$ for $n \geq 1$ by Proposition 4, while $A_0 = 1$. By Lemma 4, every $\pi \in B_n$ whose absolute fixed-point set has size $n-m$ contributes $M_m(q)$, and the fixed set may be chosen in $\binom{n}{m}$ ways; hence

$$A_n(q) = \sum_{m=0}^{n} \binom{n}{m} M_m(q).$$

Binomial inversion gives $M_n = \sum_{m=0}^{n} \binom{n}{m}(-1)^{n-m} A_m = (-1)^n A_0 = (-1)^n$. □

Proposition 5 (the vanishing half)**.** *For every $n \geq 2$, let $t \in B_n$ be the value transposition swapping ± 1 and ± 2. Right multiplication $\pi \mapsto \pi t$ is a fixed-point-free involution of the half $H = \{\pi \in B_n : \pi(1) < 0\}$; it preserves exc_B and negates both sgn_B and sgn_B^*. Hence*

$$\sum_{\pi \in H} \operatorname{sgn}_B(\pi)\, q^{\operatorname{exc}_B(\pi)} = \sum_{\pi \in H} \operatorname{sgn}_B^*(\pi)\, q^{\operatorname{exc}_B(\pi)} = 0.$$

Proof. H is closed under $\pi \mapsto \pi t$, since a value transposition maps negative values to negative values: $(\pi t)(1) = t(\pi(1)) < 0$. The map is an involution ($t^2 = e$), and $\pi t = \pi$ would force $t(\pi(i)) = \pi(i)$ for every i, hence $t = e$ since π is onto. For the excedances: right multiplication by t changes only the values $\pm 1, \pm 2$, and exc_B is a sum of per-position indicators $[\pi(i) > i]$. The values $-1, -2$ exceed no position; $+1$ exceeds no position either ($1 > i$ fails for all $i \geq 1$); and $+2$

exceeds a position i only when $i = 1$ — but position 1 holds $\pi(1) < 0$, so $+2$ sits at a position ≥ 2 both before and after the swap. Hence every indicator is unchanged and $\mathrm{exc}_B(\pi t) = \mathrm{exc}_B(\pi)$. The element t is the simple reflection s_1, so $l_B(t) = 1$ and $\mathrm{sgn}_B(t) = -1$; and since t preserves the sign of every value, $\chi_{\mathrm{neg}}(t) = +1$. Both characters being multiplicative, $\mathrm{sgn}_B(\pi t) = -\mathrm{sgn}_B(\pi)$ and $\mathrm{sgn}_B^*(\pi t) = -\mathrm{sgn}_B^*(\pi)$. Each t-orbit pairs equal-excedance permutations of opposite sign, and each sum vanishes. □

Remark 8 (the positive half). Proposition 5 cancels the negative half bijectively, and Theorem 3(i) evaluates the positive half: the entire twisted sum (11) is carried by $\{\pi(1) > 0\}$, where it equals $2(1-q)^{n-1}$. The two arguments are complementary: the involution explains *why* the negative half vanishes, the determinant *how much* the positive half is worth.

Remark 9 (type D). Let $W(D_n) = \{\pi \in B_n : \mathrm{neg}(\pi) \text{ even}\}$; the classes neg $\equiv 0, 2$ have $\chi_{\mathrm{neg}} = +1$, so $\tau = \mathrm{sgn}_B$ there, and Theorem 3(ii) gives

$$\sum_{\pi \in W(D_n)} \mathrm{sgn}_B^*(\pi)\, q^{\mathrm{exc}_B(\pi)} = \sum_{\pi \in W(D_n)} \mathrm{sgn}_B(\pi)\, q^{\mathrm{exc}_B(\pi)} = (1-q)^{n-1},$$

the type-A value rather than twice it (this was found by exact computation, before Theorem 3; see `verify.py`). An involution proving it directly inside $W(D_n)$ — a sibling of Problem 1 — remains open.

Problem 1 (bijective proof of the twisted identities)**.** Theorem 3 is algebraic: it rests on the absolute-sign identity (Lemma 3) and a two-line determinant evaluation. Find a bijective proof in the spirit of Désarménien–Foata: a τ-reversing, exc_B-preserving involution on B_n whose fixed points, refined by $\mathrm{neg}(\pi)$ mod 4, realize the four-class enumerator of Theorem 3(ii). Write δ_a for the value negation $\pi \mapsto \pi\delta_a$, interchanging the values a and $-a$ and fixing everything else. Three approaches have been investigated and found insufficient: (1) the moves δ_a never flip τ — each δ_a is a reflection of B_n, so $\mathrm{sgn}_B(\delta_a) = \chi_{\mathrm{neg}}(\delta_a) = -1$ and $\tau(\delta_a) = +1$; (2) single-pass greedy rules over the exc-preserving value-swap vocabulary (lexicographic, adjacent-first, by maximal moved value, ...) are never involutive, from $n = 3$ on (reproduced by the accompanying script); (3) of the right multiplications exhibited above, δ_1 preserves exc_B on all of B_n (Proposition 4) and $t = (1\,2)$ preserves it on the half $\{\pi : \pi(1) < 0\}$ (Proposition 5), and neither extends to a full-group fixed-point enumeration. We have no classification of the right multiplications that preserve exc_B; such a classification would be a first step. The naive generalization of the latter also fails: on $\{\pi : \pi(c), \pi(c+1) < 0\}$, right multiplication by $t_c = (c\ c+1)$ preserves exc_B only when the positions of $+c$ and $+(c+1)$ lie on the same side of c, which is automatic for $c = 1$ (Proposition 5) but not for $c \geq 2$. The proof of Theorem 1 does not transfer because $(-1)^{l_B}$ contains the pair term $\#\{i \leq j : \sigma(i) + \sigma(j) < 0\}$, which is not entry-wise; Lemma 3 shows this term is benign modulo 2, and a combinatorial model tracking it exactly is what a bijective proof requires.

The closest prior art is Sivasubramanian's signed excedance enumeration of the hyperoctahedral group. In [16] he weights B_n by $(-1)^{l_B} = \mathrm{sgn}_B$ but with Brenti's excedance, which counts negative fixed points: his full-group sum is $(1-q)^n$ and his derangement sum is $(-q)^n$, both different from the values of Proposition 4 and Theorem 5 because the statistics differ. In [17] he enumerates B_n against all four one-dimensional characters — noting explicitly that $(-1)^{l_B + \mathrm{neg}} = \mathrm{sgn}(|\pi|)$, the twist of Lemma 3 — but again with Brenti's excedance, obtaining $\sum_{B_n} \tau(\pi)\, q^{\mathrm{exc}_{\mathrm{Brenti}}(\pi)} = (1+q)(1-q)^{n-1}$ for odd n and $(1-q)^n$ for even n; the two answers differ from Theorem 3(i) because the statistics do. The same Brenti excedance appears with the type-B cycle-count character in [21], whose signed polynomial $\sum_{B_n} (-1)^{\mathrm{cyc}(\pi)} x^{\mathrm{exc}_{\mathrm{Brenti}}(\pi)}$ equals $-(x+1)(x-1)^{n-1}$ for odd n and $(x-1)^n$ for even n [21, Thm. 3], and signed excedance

enumeration of classical and affine Weyl groups is the subject of [10], which we could not access; we cannot exclude that (11) appears there. With the plain absolute excedance and the four characters $1, \mathrm{sgn}_B, \chi_{\mathrm{neg}}, \tau$, Theorem 3 and its four-class refinement match nothing else we could verify; the reformulation $\tau = \mathrm{sgn}(|\pi|)$ (Lemma 3) should make the comparison easy. See Problem 4.

5. A graded Alon–Tarsi polynomial

The programme sketched in Section 1 — grade a signed enumeration by distance from a canonical object — can be carried out for the Alon–Tarsi constant [1] itself, and the derangement machinery of Sections 3–4 governs its lowest layers. Fix the cyclic Latin square C of order n, $C(i,j) = ((i+j-2) \bmod n) + 1$. For a Latin square L put

$$\mathrm{sgn}(L) := \prod_{i=1}^{n} \mathrm{sgn}\big(L(i,\cdot)\big) \prod_{j=1}^{n} \mathrm{sgn}\big(L(\cdot,j)\big),$$

the product of the signs of its n rows and its n columns, each read as a permutation, and let $d(L,C)$ be the number of cells in which L differs from C. Define

$$\mathrm{AT}_n^{\mathrm{cyc}}(q) := \sum_{L} \mathrm{sgn}(L)\, q^{d(L,C)}, \tag{15}$$

the sum over all Latin squares L of order n; thus $\mathrm{AT}_n^{\mathrm{cyc}}(1) = \mathrm{AT}_n$ is the Alon–Tarsi constant, the difference between the numbers of even and odd Latin squares. Exact computation gives

$$\mathrm{AT}_3^{\mathrm{cyc}}(q) = 1 - 3q^6 + 2q^9,$$

$$\mathrm{AT}_4^{\mathrm{cyc}}(q) = 1 + 4q^4 + 32q^7 + 18q^8 + 96q^{10} + 204q^{12} + 64q^{13} + 96q^{14} + 32q^{15} + 29q^{16},$$

with $\mathrm{AT}_3^{\mathrm{cyc}}(1) = 0$ (the known vanishing for odd orders) and $\mathrm{AT}_4^{\mathrm{cyc}}(1) = 576$. Two features invite study: the base layer is $+1$ — the cyclic square is even and, for $n = 3$, isolated in its layer, a direct Latin-square analogue of the unique-uncancelled- derangement phenomenon of Theorem 2 — and the odd-order vanishing at $n = 3$ is achieved by cancellation *across* layers, not layer by layer. We note also that the naive graded layers of the Vandermonde square trivialize. Writing $\Delta = \prod_{i<j}(x_i - x_j) = \det\big(x_i^{n-j}\big)_{i,j=1}^{n}$, one has $\prod_{i\neq j}(x_i - x_j) = (-1)^{\binom{n}{2}}\Delta^2$, and the coefficient of $\prod_i x_i^{n-1}$ is $n!$ in $\prod_{i\neq j}(x_i - x_j)$ and $(-1)^{\binom{n}{2}} n!$ in Δ^2: for each $\sigma \in S_n$ there is exactly one $\rho \in S_n$ with $\sigma(i) + \rho(i) = n+1$ for all i, namely $\rho = w_0\sigma$, and $\mathrm{sgn}(\sigma)\,\mathrm{sgn}(\rho) = (-1)^{\binom{n}{2}}$. Hence the graded program must engage the Latin-square structure and cannot proceed by graph-polynomial layers alone.

A fully rigorous grading is available through the determinant-sum formula for the Alon–Tarsi constant, due to Glynn [7] and surveyed among the many Latin-rectangle formulae of [18]:

$$\mathrm{AT}_n = \sum_{A \in \{0,1\}^{n\times n}} (-1)^{|A|} \det(A)^n, \tag{16}$$

where $|A|$ is the number of ones. Grading (16) by sparsity,

$$\mathrm{AT}_n^{\mathrm{sp}}(q) := \sum_{A} (-1)^{|A|} \det(A)^n\, q^{|A|} = \sum_{k} L_{n,k}\, q^k, \tag{17}$$

gives a genuinely graded polynomial, whose support Proposition 6 below places in $[\,n,\ n^2-n+1\,]$; for $n = 4$,

$$\mathrm{AT}_4^{\mathrm{sp}}(q) = 24\big(q^4 - 12q^5 + 60q^6 - 216q^7 + 582q^8 - 912q^9 + 684q^{10} - 264q^{11} + 105q^{12} - 4q^{13}\big).$$

Three phenomena are visible in this polynomial. The first two are proved below, for all n, by a single orbit computation (Theorem 6); the third is Corollary 8.

(a) *Layer-wise vanishing for odd orders*: every $L_{n,k}$ is zero when $n \geq 3$ is odd — the graded refinement of $\mathrm{AT}_n = 0$.
(b) *Divisibility by $n!$ for even orders*: every $L_{n,k}$ is a multiple of $n!$ when n is even.
(c) *Strict sign alternation for even orders*: on every level admitting a nonsingular matrix, $L_{n,k}$ has sign $(-1)^k$.

Theorem 6 (layer structure of the graded Alon–Tarsi polynomial)**.** *For every $n \geq 2$ and every sparsity level k:*

$$L_{n,k} := \sum_{|A|=k} (-1)^{|A|} \det(A)^n = \begin{cases} n! \sum_{O} \det(A_O)^n & n \text{ even}, \\ 0 & n \text{ odd}, \end{cases}$$

where O runs over the orbits of the row-permutation action $S_n \curvearrowright \{A : |A| = k\}$ and $A_O \in O$ is any representative. In particular: (i) for odd $n \geq 3$ every layer vanishes identically — the graded, layer-wise refinement of the classical vanishing $\mathrm{AT}_n = 0$ at odd orders; (ii) for even n every layer coefficient is divisible by $n!$.

Proof. S_n acts on $n \times n$ matrices by permuting rows; the action preserves the 0-1 property, the number of ones, and hence each sparsity layer. On an orbit O with representative A_O and any $\rho \in S_n$ we have $\det(\rho A_O)^n = \mathrm{sgn}(\rho)^n \det(A_O)^n$, and the map $\rho \mapsto \rho A_O$ is $|\mathrm{Stab}(A_O)|$-to-one onto O, so

$$\sum_{B \in O} \det(B)^n = \frac{\det(A_O)^n}{|\mathrm{Stab}(A_O)|} \sum_{\rho \in S_n} \mathrm{sgn}(\rho)^n.$$

If n is even then $\mathrm{sgn}(\rho)^n = 1$ and the right-hand side is $|O| \det(A_O)^n$; moreover a nontrivial stabilizer forces two rows of A_O to be equal — $\rho A_O = A_O$ with $\rho \neq e$ means the rows i and $\rho^{-1}(i)$ coincide for any i moved by ρ — hence $\det(A_O) = 0$, so $|O| = n!$ unless the contribution vanishes anyway. Either way the orbit contributes $n! \det(A_O)^n$. If $n \geq 3$ is odd then $\mathrm{sgn}(\rho)^n = \mathrm{sgn}(\rho)$ and $\sum_{\rho \in S_n} \mathrm{sgn}(\rho) = 0$, so the orbit contributes 0. Summing over orbits proves both statements. □

Remark 10. The restriction $n \geq 3$ in the odd case is necessary: for $n = 1$ the group S_1 has no sign-cancelling pair and $L_{1,1} = -1$.

Corollary 8 (strict sign alternation for even orders)**.** *For even n, $L_{n,k}$ vanishes outside $[\, n,\ n^2 - n + 1 \,]$, and on every sparsity level that admits a nonsingular 0-1 matrix,*

$$L_{n,k} = (-1)^k \sum_{|A|=k} |\det A|^n = (-1)^k \, n! \, \widehat{R}_{n,k}, \qquad \widehat{R}_{n,k} > 0,$$

where $\widehat{R}_{n,k}$ is an integer: a count of row-permutation orbits weighted by $|\det|^n$, whose stabilizers are all trivial because a matrix with $\det \neq 0$ has distinct rows. For $n \leq 4$ every level of $[\, n,\ n^2 - n + 1 \,]$ admits such a matrix (verified exhaustively; see Proposition 6), so throughout the support the layer signs strictly alternate with the sparsity k.

Proof. For even n, $\det(A)^n = |\det A|^n \geq 0$, with strict positivity exactly on the nonsingular matrices. By Theorem 6, each orbit with nonsingular representative contributes $n! \det(A_O)^n$ and each singular orbit contributes 0; summing over orbits gives the displayed identity with $\widehat{R}_{n,k} = \sum_O |\det(A_O)|^n > 0$ whenever a nonsingular representative exists. Support containment is Proposition 6. □

For $n = 4$ the sequence $\widehat{R}_{4,k}$, $k = 4, \dots, 13$, is 1, 12, 60, 216, 582, 912, 684, 264, 105, 4; we have not found it in the OEIS.

Consequently the entire odd-order vanishing of the Alon–Tarsi constant is witnessed *layer by layer* in the sparsity grading (17), and for even orders the constant is $n!$ times an integer. The orbit sums for even n are signed counts of row-permutation orbits whose representatives have full rank — the low orbits are precisely the "almost permutation matrices" computed in Corollary 9.

Proposition 6 (support and endpoints)**.** *The support of* $\mathrm{AT}_n^{\mathrm{sp}}(q)$ *is contained in* $[\,n,\ n^2-n+1\,]$*, and both endpoints are attained:* I_n *at* $k = n$*, and* $J_n - P + e_{12}$ *at* $k = n^2 - n + 1$*, where* J_n *is the all-ones matrix,* e_{12} *the matrix unit in position* $(1, 2)$*, and* P *the cyclic shift matrix* $P = \sum_{i=1}^{n} e_{i,i+1}$ *with* $e_{n,n+1} := e_{n,1}$*. The matrix* $J_n - P + e_{12}$ *is nonsingular for every* $n \geq 2$*. Exhaustive computation gives support exactly* $[3, 7]$ *for* $n = 3$ *and* $[4, 13]$ *for* $n = 4$*; a uniform construction of nonsingular representatives across the whole interior range remains open.*

Proof. A 0-1 matrix with fewer than n ones has an all-zero row, hence vanishing determinant. A 0-1 matrix with more than $n^2 - n + 1$ ones has fewer than $n - 1$ zeros, so at least two of its rows are all ones, and two equal rows force the determinant to vanish. Both endpoints are attained: $\det I_n = 1$ at $k = n$; and for $A = J_n - P + e_{12}$ (all ones except the $n - 1$ cells $(i, i+1)$, $2 \leq i \leq n$, indices taken cyclically) we have $|A| = n^2 - n + 1$, and $Ax = 0$ reads $S = \sum_j x_j = 0$ on row 1 and $S - x_{i+1} = 0$ on rows $i \geq 2$ (with $x_{n+1} := x_1$), whence $x_3 = \cdots = x_n = x_1 = 0$ and $S = 0$ forces $x_2 = 0$: A is nonsingular. □

The bottom layers are exactly where the derangement machinery of this note operates, and the first two are computed exactly.

Lemma 5 (one-cell invariance)**.** *Let* P_π *be a permutation matrix,* $i \in [n]$*, and* $j \neq \pi(i)$*. Then* $\det(P_\pi + e_{ij}) = \det(P_\pi) = \mathrm{sgn}(\pi)$.

Proof. Expanding along row i, the original one at column $\pi(i)$ contributes $\mathrm{sgn}(\pi)$; the added cell contributes $(-1)^{i+j} \det(M)$, where M is P_π with row i and column j deleted. Since $j \neq \pi(i)$, the column $\pi(i)$ of M is empty (its single 1 sat in row i), so $\det M = 0$. □

Corollary 9 (the first two layers)**.**

$$L_{n,n} = \sum_{|A|=n} (-1)^{|A|} \det(A)^n = \begin{cases} n! & n \text{ even} \\ 0 & n \text{ odd,} \end{cases}$$

$$L_{n,n+1} = \sum_{|A|=n+1} (-1)^{|A|} \det(A)^n = \begin{cases} (-1)^{n+1}\, n(n-1)\, n! & n \text{ even} \\ 0 & n \text{ odd.} \end{cases}$$

Both match the $n = 3, 4$ *data above. Note* $L_{4,4} = 24$ *and* $L_{4,5} = -288 = -4 \cdot 3 \cdot 24$.

Proof. A 0-1 matrix of size n with $\det \neq 0$ must contain a permutation matrix. For $|A| = n$ it *is* one, giving $L_{n,n} = (-1)^n \sum_\pi \mathrm{sgn}(\pi)^n$. For $|A| = n + 1$: a matrix with $n + 1$ ones contains at most one permutation matrix (two permutation matrices whose ones differ in exactly two cells would have a union of size $n + 2$; differing by a transposition, the union matrix is block diagonal with an all-ones 2×2 block, hence determinant 0). Each admissible A is thus uniquely $P_\pi + e_{ij}$, $j \neq \pi(i)$, and by Lemma 5 $\det A = \mathrm{sgn}(\pi)$; there are $n(n-1)$ choices of (i, j) per π. So $L_{n,n+1} = (-1)^{n+1} n(n-1) \sum_\pi \mathrm{sgn}(\pi)^n$. For odd $n \geq 3$, $\mathrm{sgn}(\pi)^n = \mathrm{sgn}(\pi)$ and $\sum_\pi \mathrm{sgn}(\pi) = 0$; for even n, $\mathrm{sgn}(\pi)^n = 1$. □

Experimentally (n=4), layer $|A| = 6$ is also carried entirely by matrices whose support contains exactly one permutation matrix (total 1440; the two-permutation unions all have determinant

0, their transposition-type unions being block diagonal with an all-ones 2×2 block), and the general-n closed form of layer $|A| = n + 2$ involves cross-swap signs and is left open. A uniform evaluation of the low layers would be the first concrete step of the graded program sketched above.

6. Concluding remarks and problems

Remark 11 (Lucas-type form). The recurrence (6) exhibits $D_n(y, q) = \det T_n(y, q)$ as a Lucas-type sequence in n with characteristic values $y - 1$ and $y - q$. All coefficients of $F_n(q)$ are ± 1 (Theorem 2), which makes it usable as an exact benchmark for programs that compute signed enumerators.

Problem 2 (bijective proof)**.** For each $n \geq 2$ and $1 \leq k \leq n - 1$, construct a sign-reversing involution on $\{\pi \in \mathcal{D}_n : \mathrm{exc}(\pi) = k\}$ having exactly one unpaired derangement. The involution of [3] proves Corollary 3 but does not preserve exc. Such an involution does exist: the killing involution φ_1 of [8, §2.2] preserves exc and reverses $(-1)^{\mathrm{cyc}}$ on $\mathcal{D}_n \setminus (U_n \cup \{\sigma_0\})$, where $U_n = \{\sigma \in \mathcal{D}_n : \sigma(2) = 1,\ \sigma(1) \neq 2\}$ and $\sigma_0 = (1, 2, \ldots, n)$, and unwound through their recursion $P_n(x, 0, -1) = P_{n-1}(x, 0, -1) - x^{n-1}$ it leaves exactly one survivor per level — σ_0 at level $n - 1$, and below it the pull-back, through the weight-preserving bijection $U_n \to \mathcal{D}_{n-1}$, of the survivor of $\mathcal{D}_{n-1}$. Their Theorem 1 refines (4) by the last value $\sigma(n) = j$ rather than by the level. What remains open is a *direct* involution, one that exhibits the survivor of level k without recursion; the signed Eulerian identity (3) underlying Theorem 2 does admit a bijective proof by a sign-reversing involution [20], so what is missing is precisely a combinatorial model for the passage from S_n to $\mathcal{D}_n$ that keeps track of exc and sgn simultaneously.

Two natural attacks fail, which delimits the difficulty. The most tempting rule — swap the adjacent positions $i, i + 1$ at the *first* i where the swap preserves exc (equivalently, where $[\pi(i) = i + 1] = [\pi(i + 1) = i + 1]$) — does preserve exc, flip the sign, and every orbit is finite; but it is an involution only for $n = 2$, and from $n = 3$ on its orbits close into cycles of length > 2, e.g.

$$(3, 2, 1) \mapsto (3, 1, 2) \mapsto (1, 3, 2) \mapsto (3, 2, 1).$$

The derangement-restricted variant (additionally requiring $\pi(i) \neq i + 1$ and $\pi(i + 1) \neq i$, so that derangements map to derangements) behaves the same way, failing from $n = 4$ on: $(3, 1, 4, 2) \mapsto (3, 4, 1, 2) \mapsto (4, 3, 1, 2) \mapsto (3, 4, 1, 2) \mapsto \cdots$ (the orbit falls into the 2-cycle of another pair). Moreover, several derangements are sent to unpaired elements outright, e.g. $(2, 3, 4, 1) \mapsto \emptyset$ and $(2, 4, 1, 3) \mapsto \emptyset$ for $n = 4$. The construction thus seems to require the marked-permutation technique of [6] rather than a local rule; the failing orbits are reproduced by the accompanying script. A homological reformulation may be the right frame: by Corollary 2 the free abelian group on each level $\{\pi \in \mathcal{D}_n : \mathrm{exc}(\pi) = k\}$ has Euler characteristic $(-1)^{n-1}$ under any sign-reversing differential, so the problem asks for a natural cell structure on each level whose reduced homology is one-dimensional, concentrated at the survivor. Computationally (Tutte-matrix matching tests, $n \leq 6$), the pairing graph on each level admits near-perfect matchings leaving *any* prescribed majority-sign derangement as the survivor: Problem 2 is a canonization problem, not an existence problem (the script `experiment_survivors.py` reproduces this).

Problem 3 ($q = -1$ involution, even case)**.** For odd n, Proposition 2 settles Corollary 7 bijectively via the reverse-complement. The cancellation of the non-reverse-complementary layer is settled by Proposition 3 together with Corollary 7, and the weight $+1$ of the reverse-complementary layer is evaluated directly in the proof of Proposition 3, by inclusion–exclusion over $\mathcal{D}(B_m)$ (these permutations are determined by their first $n/2$ entries; see also [11]). What remains open is an explicit weight-reversing involution on the non-reverse-complementary derangements. The natural moves — right-multiplying by the value-negations δ_a (swapping the values $\pm a$)

— preserve exc_B only when $a \le \pi^{-1}(\pm a)$ for the two affected positions (for $a = 1$ this always holds, which is what drives Proposition 4), so an exc-preserving weight-reversing involution must combine several such moves. The signed Eulerian numbers, i.e. the $q = -1$ evaluations of the Eulerian polynomials, are the subject of [6] and of its bijective sequel [20]; whether their methods reach the derangement fiber (8) at $q = -1$ is not clear to us.

Problem 4 (statistics for type B and beyond)**.** Carry Problem 1 out, and extend the determinant method beyond the excedance statistic. Experimentally, with $\mathrm{inv}(\pi) = \#\{i < j : \pi(i) > \pi(j)\}$ the ordinary inversion count of the signed one-line word,

$$\sum_{\pi \in B_n} \tau(\pi)\, x^{\mathrm{neg}(\pi)}\, q^{\mathrm{inv}(\pi)} = (1+x)^{n \bmod 2}\,(1-x^2)^{\lfloor n/2 \rfloor} \prod_{j=1}^{n-1} \bigl(1 - q + q^2 - \cdots + (-q)^j\bigr)$$

(verified $n \le 6$; note that inv is *not* entry-wise, so this does not follow from Theorem 3 — prove it). Specializing to $x = 0$ isolates the negation-free part and gives $\sum_{w \in S_n} \mathrm{sgn}(w) q^{\mathrm{inv}(w)} = \prod_{j=1}^{n-1}(1 - q + \cdots + (-q)^j)$, the Désarménien–Foata identity (3); the content is therefore entirely in the x-direction. Beyond type B, ask for the analogue of Theorem 3 for the colored groups of [2], and for the absolute derangement set $\mathcal{D}(B_n)$ replaced by other type-B derangement conventions.

Remark 12 (scope). Toeplitz determinants of the shape $T_n(y, q)$ are classical objects, and the evaluation in Theorem 1 presumably exists in the linear-algebra literature. The present note claims novelty only for the enumerative packaging — the simultaneous refinement by fixed points and excedances — and for the rigidity statement of Corollary 2, for which no prior reference is known to the author. All enumerative data quoted in this paper were verified by exact computation, in the ranges listed in Appendix A.

Lemma 6 (the extreme levels are signed long cycles)**.** *For $n \ge 2$ write $c_n = (n, 1, 2, \ldots, n-1) \in S_n$ for the long cycle in one-line notation, and for $1 \le k \le n-1$ let*

$$\varepsilon_{n,k} = \sum_{\pi \in \mathcal{D}_n,\ \mathrm{exc}(\pi) = k} \mathrm{sgn}(\pi)\, \pi \ \in\ \mathbb{Q}S_n$$

be the level element *of the k-th signed excedance layer (not to be confused with the even-derangement counts $E_{n,k}$ of Corollary 2). Then c_n is the only derangement with a single excedance and c_n^{-1} the only one with $n-1$ excedances; in particular*

$$\varepsilon_{n,1} = (-1)^{n-1} c_n, \qquad \varepsilon_{n,n-1} = (-1)^{n-1} c_n^{-1} = \varepsilon_{n,1}^{-1}, \qquad \varepsilon_{n,1}^{\,n} = 1,$$

so the two extreme layers of $F_n(q)$ consist of a single group element each, carrying the same sign $(-1)^{n-1}$.

Proof. Let $\pi \in \mathcal{D}_n$ have its unique excedance at position i. Then $\pi(j) < j$ for every $j \ne i$, and since $\pi(1) < 1$ is impossible, $i = 1$. Now $\pi(j) \le j - 1$ for $2 \le j \le n$ while $\pi(2), \ldots, \pi(n)$ are distinct, so $\pi(2) = 1$, then $\pi(3) = 2$, and inductively $\pi(j) = j - 1$; hence $\pi(1) = n$ and $\pi = c_n$. The statement for $n-1$ excedances is the mirror argument: the unique non-excedance must sit at n, and then $\pi(j) = j + 1$ for $j < n$, so $\pi = c_n^{-1}$. Both permutations are n-cycles, of sign $(-1)^{n-1}$, and $\varepsilon_{n,1}^{\,n} = (-1)^{n(n-1)} c_n^{\,n} = 1$ because $n(n-1)$ is even. □

The next four results are unconditional. Together they explain part of the data assembled in Remark 15 and reduce Problem 5(a) to a single concrete statement about coefficients.

Lemma 7 (inversion is the reverse-complement on the levels)**.** *Let $\iota \colon \mathbb{Q}S_n \to \mathbb{Q}S_n$ be the linear anti-involution $\pi \mapsto \pi^{-1}$ and let w_0 be the reversal $i \mapsto n + 1 - i$, so that $\bar{\pi} = w_0 \pi w_0$ is the reverse-complement of Proposition 2. Then for $1 \le k \le n-1$*

$$\iota(\varepsilon_{n,k}) \ =\ \varepsilon_{n,n-k} \ =\ w_0 \varepsilon_{n,k} w_0, \tag{18}$$

hence $\iota(\varepsilon(q)) = q^n\varepsilon(1/q)$ *and* $\iota(A_n) = w_0A_nw_0 = A_n$. *The composite* $\rho = \mathrm{Ad}(w_0)\circ\iota$, *that is* $\rho(\pi) = w_0\pi^{-1}w_0 = \overline{\pi^{-1}}$, *is an involution of* S_n *which preserves each level set* $\{\pi\in\mathcal{D}_n : \mathrm{exc}(\pi) = k\}$ *and the sign on it; in particular* $\rho(\varepsilon_{n,k}) = \varepsilon_{n,k}$ *and* $\rho(A_n) = A_n$.

Proof. For $\pi\in\mathcal{D}_n$, $\mathrm{exc}(\pi^{-1}) = \#\{i : \pi(i) < i\} = n - \mathrm{exc}(\pi)$ because π has no fixed point, and $\mathrm{exc}(w_0\pi w_0) = n - \mathrm{exc}(\pi)$ by Proposition 2; both maps are bijections of $\mathcal{D}_n$ and both preserve sgn. Substituting $\sigma = \pi^{-1}$, respectively $\sigma = w_0\pi w_0$, in the sum defining $\varepsilon_{n,k}$ gives (18); the graded identity is the re-indexing $k\mapsto n-k$, legal because a derangement has between 1 and $n-1$ excedances. Since $\rho^2(\pi) = w_0(w_0\pi w_0)w_0 = \pi$, ρ is an involution, and it preserves levels and signs because each factor does; so it fixes every $\varepsilon_{n,k}$, and being an anti-automorphism of $\mathbb{Q}S_n$ that maps the generating set of A_n into A_n it maps A_n into itself. □

Proposition 7 (a criterion for commutativity, and semisimplicity)**.** *Let* $n\geq 3$.

(i) A_n *is commutative if and only if* $\iota(x) = w_0xw_0$ *for every* $x\in A_n$, *that is, if and only if* $\rho(x) = x$ *for every* $x\in A_n$. *Equivalently, writing* $x = \sum_\pi x_\pi\pi$, *commutativity is the statement that* $x_\pi = x_{\rho(\pi)}$ *for every* $\pi\in S_n$ *and every* $x\in A_n$: *the generators satisfy this coefficient symmetry by Lemma 7, and the content of the criterion is the question whether multiplication preserves it.*

(ii) A_n *is semisimple, for every* n *and without any commutativity hypothesis. If* A_n *is commutative then* $A_n\otimes_{\mathbb{Q}}\mathbb{C}\cong\mathbb{C}^{\dim A_n}$, A_n *has exactly* $\dim A_n$ *characters, and each of them satisfies*

$$\chi(\iota(x)) = \overline{\chi(x)}\qquad (x\in A_n), \tag{19}$$

so that $\chi(\varepsilon_{n,n-k}) = \overline{\chi(\varepsilon_{n,k})}$ *and* $\chi(\varepsilon(q)) = q^n\overline{\chi(\varepsilon(1/q))}$ *for real* $q\neq 0$. *The character table of* A_n *is therefore palindromic in the level index* $k\leftrightarrow n-k$, *and* $\det(x - L_{\varepsilon(q)})$ *has real coefficients with* $\mathrm{spec}\,L_{\varepsilon(1/q)} = q^{-n}\overline{\mathrm{spec}\,L_{\varepsilon(q)}}$.

(iii) $\mathrm{Spec}\,A_n$ *is stable under* $\mathrm{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})$, *which acts by* $\chi\mapsto\chi^g$, $\chi^g(x) = g(\chi(x))$. *If* m_j *denotes the number of characters with* $\chi(c_n) = \zeta_n^j$, *then* $m_j = m_{gj}$ *whenever* $\gcd(g,n) = 1$; *for* n *prime this forces* $m_1 = \cdots = m_{n-1} = (\dim A_n - m_0)/(n-1)$ *and* $\dim A_n\equiv m_0 \pmod{n-1}$.

Proof. (i) If $\iota = \mathrm{Ad}(w_0)$ on A_n then for $x,y\in A_n$, $\iota(xy) = \iota(y)\iota(x) = w_0yw_0\,w_0xw_0 = w_0yxw_0 = \iota(yx)$, and ι is injective. Conversely, if A_n is commutative then ι, an anti-automorphism, restricts to an automorphism of A_n, as does $\mathrm{Ad}(w_0)$, and by (18) the two automorphisms agree on 1 and on every generator $\varepsilon_{n,k}$, hence on A_n. The coefficient formulation is $\rho(x) = x \iff x_\pi = x_{\rho(\pi)}$ for all π. (For $x,y\in A_n$ fixed by ρ one has $(xy)^\rho = y^\rho x^\rho = yx$, so the symmetry does propagate through a product precisely when the two factors commute — which is why the criterion is an equivalence rather than a proof.) (ii) On $\mathbb{C}S_n$ let $x^* = \sum_\pi\bar{x}_\pi\pi^{-1}$ (conjugate linear) and $\langle x,y\rangle = \sum_\pi\bar{x}_\pi y_\pi$; the left regular representation λ is a faithful $*$-representation, since permutation matrices are unitary, and $\langle x,y\rangle = \frac{1}{n!}\,\mathrm{tr}\,\lambda(x^*y)$, so $\langle x,x\rangle = 0$ forces $x = 0$. By Lemma 7, $A_n\otimes\mathbb{C}$ is $*$-stable: ι preserves A_n and complex conjugation preserves it because A_n is defined over $\mathbb{Q}$. If J were a nonzero nilpotent ideal of $A_n\otimes\mathbb{C}$, then J is ι-stable ($\iota(J)$ is again a nilpotent ideal) and conjugation-stable, hence $*$-stable, and for $0\neq x\in J$ the element $x^*x\in J$ would be nilpotent, giving $0 < \langle x,x\rangle = \frac{1}{n!}\,\mathrm{tr}\,\lambda(x^*x) = 0$; so $J = 0$ and $A_n\otimes\mathbb{C}$, hence A_n, is semisimple. A semisimple commutative finite-dimensional algebra over $\mathbb{C}$ is a product of copies of $\mathbb{C}$, one per character, and being a $*$-subalgebra of the C*-algebra $\mathbb{C}S_n$ it is a C*-algebra itself, so the product carries componentwise conjugation and every character is a $*$-homomorphism: $\chi(x^*) = \overline{\chi(x)}$. For $x\in A_n\subseteq\mathbb{Q}S_n$ we have $x^* = \iota(x)$, which is (19); the two spectral statements follow by applying it to $x = \varepsilon(q)$ and using $\iota(\varepsilon(q)) = q^n\varepsilon(1/q)$. (iii) The eigenvalues of L_x are algebraic for every $x\in A_n$, so every character takes values in $\bar{\mathbb{Q}}$, and χ^g is again a $\mathbb{Q}$-algebra homomorphism

$A_n \to \mathbb{C}$. Since $\chi^g(c_n) = g(\zeta_n^j) = \zeta_n^{gj}$, the bijection $\chi \mapsto \chi^g$ carries $\{\chi : \chi(c_n) = \zeta_n^j\}$ onto $\{\chi : \chi(c_n) = \zeta_n^{gj}\}$, so $m_j = m_{gj}$. For n prime the group $(\mathbb{Z}/n)^\times$ acts transitively on $\{1, \dots, n-1\}$, and $\dim A_n = \sum_j m_j$. □

Remark 13. Proposition 7(ii) upgrades two pieces of computational evidence in Remark 15(i) to theorems: the trace form of A_n is not merely nondegenerate for $n \le 6$ but the restriction of a positive definite Hermitian form for every n, so A_n is étale over $\mathbb{Q}$ whenever it is commutative; and the palindromy of the computed character tables is forced rather than observed. Part (iii) explains the shape of the computed $\mathbb{Q}C_n$-multiplicities: the values of m_j must be constant on the orbits of $(\mathbb{Z}/n)^\times$, which for $n = 4$ predicts $m_1 = m_3$ and for $n = 6$ predicts $m_1 = m_5$, $m_2 = m_4$, exactly as computed, while for prime n it predicts a single value off $j = 0$.

Proposition 8 (the two evident characters: $F_n(q)$ and $d_n(q)$)**.** *The augmentation* $\mathrm{aug}\big(\sum_\pi a_\pi \pi\big) = \sum_\pi a_\pi$ *and the sign* $\mathrm{sgn}\colon \mathbb{Q}S_n \to \mathbb{Q}$ *are algebra homomorphisms, so they restrict to characters of* A_n*, with*

$$\mathrm{aug}\big(\varepsilon(q)\big) = F_n(q), \qquad \mathrm{sgn}\big(\varepsilon(q)\big) = d_n(q) := \sum_{k=1}^{n-1} d_{n,k} q^k, \tag{20}$$

where $F_n(q)$ *is the signed enumerator of Theorem 2 and* $d_n(q)$ *the derangement Eulerian polynomial of Remark 3. Thus the signed and the unsigned derangement enumerators are two eigenvalues of one integral matrix* $L_{\varepsilon(q)}$*, and both are roots of* $\det(x - L_{\varepsilon(q)})$*, a monic polynomial of degree* $\dim A_n$ *with integer coefficients. The two characters coincide only for* $n = 3$*, where* $d_{3,1} = d_{3,2} = 1$*; for* $n \ge 4$ *they differ already at* $\varepsilon(1)$*, where they take the values* $(-1)^{n-1}(n-1)$ *and* $!n$*. They are the* only *characters of* A_n *that extend to* $\mathbb{Q}S_n$*, because* S_n *has exactly two linear characters.*

Proof. $\mathrm{aug}(\varepsilon_{n,k}) = E_{n,k} - O_{n,k} = (-1)^{n-1}$ by Corollary 2, so $\mathrm{aug}(\varepsilon(q)) = (-1)^{n-1}(q + \cdots + q^{n-1}) = F_n(q)$ by Theorem 2; and $\mathrm{sgn}(\varepsilon_{n,k}) = \sum_\pi \mathrm{sgn}(\pi)^2 = d_{n,k}$, giving $\mathrm{sgn}(\varepsilon(q)) = d_n(q)$. Both are therefore eigenvalues of $L_{\varepsilon(q)}$, with eigenvectors the primitive idempotents $e_{\mathrm{aug}}, e_{\mathrm{sgn}}$ of $A_n \otimes \mathbb{C}$ supplied by Proposition 7(ii). At $\varepsilon(1) = \sum_{\pi \in \mathcal{D}_n} \mathrm{sgn}(\pi)\pi$ the two take the values $F_n(1) = (-1)^{n-1}(n-1)$ and $d_n(1) = !n$, which are equal only for $n \le 3$. Finally an extension of a character of A_n to $\mathbb{Q}S_n$ is an algebra homomorphism $\mathbb{Q}S_n \to \mathbb{C}$, that is, a linear character of S_n, and there are only aug and sgn. □

Definition 1 (level-multiplicative characters)**.** *A character* χ *of* A_n *is* level-multiplicative *if* $\chi(\varepsilon_{n,k}) = \chi(\varepsilon_{n,1})^k$ *for* $1 \le k \le n-1$*. By Lemma 6* $\chi(\varepsilon_{n,1})^n = \chi(1) = 1$*, so such a character is one of the* χ_a*,* $a \in \mathbb{Z}/n$*, defined by*

$$\chi_a(\varepsilon_{n,k}) = \zeta_n^{ak}, \qquad \chi_a(\varepsilon(q)) = \sum_{k=1}^{n-1} (q\zeta_n^a)^k, \qquad \chi_a(c_n) = (-1)^{n-1}\zeta_n^a, \tag{21}$$

provided that the assignment is well defined on A_n*. Since* $(-1)^{n-1}F_n(q) = q + \cdots + q^{n-1}$*, the second formula says that* $\chi_a(\varepsilon(q))$ *is the image of* $(-1)^{n-1}F_n(q)$ *under the substitution* $q \mapsto q\zeta_n^a$*: the level-multiplicative eigenvalues of* $L_{\varepsilon(q)}$ *are the Galois conjugates of the augmentation eigenvalue, obtained by twisting* q *by the* n*-th roots of unity.*

Proposition 9 (the cyclotomic characters)**.** (i) *If* n *is odd then* χ_0 *is a character of* A_n *and equals* aug*. If* n *is even then* $\chi_0 \ne \mathrm{aug}$*, since* $\chi_0(\varepsilon_{n,k}) = 1$ *while* $\mathrm{aug}(\varepsilon_{n,k}) = (-1)^{n-1} = -1$.

(ii) $A_3 = \mathbb{Q}[c_3] = \mathbb{Q}C_3 \cong \mathbb{Q} \times \mathbb{Q}(\zeta_3)$, *and* $\operatorname{Spec} A_3$ *consists of exactly the three level-multiplicative characters* $\chi_0 = \mathrm{aug} = \mathrm{sgn}$, χ_1, χ_2. *Consequently*

$$\det\big(x - L_{\varepsilon(q)}\big) = \prod_{a=0}^{2}\Big(x - \textstyle\sum_k (q\zeta_3^a)^k\Big) = \big(x - F_3(q)\big)\, N_{\mathbb{Q}(\zeta_3)/\mathbb{Q}}\big(x - \textstyle\sum_k (q\zeta_3)^k\big), \tag{22}$$

which at $q = 2$ *is* $(x-6)(x^2+6x+12) = x^3 - 24x - 72$.

(iii) *If* χ_a *is a character of* A_n *and* $\gcd(g,n) = 1$ *then* $\chi_{ga} = \chi_a^{\,g}$ *is a character of* A_n, *where* $g \in \mathrm{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})$ *sends* $\zeta_n \mapsto \zeta_n^g$. *Hence* χ_1 *exists if and only if every* χ_a *with* $\gcd(a,n) = 1$ *exists, and for* n *prime this means all of* $\chi_1, \dots, \chi_{n-1}$; *those* $n-1$ *characters then form a single Galois orbit with field of values* $\mathbb{Q}(\zeta_n)$, *so* $\mathbb{Q}(\zeta_n)$ *is a direct factor of* A_n *and*

$$\prod_{a=1}^{n-1}\Big(x - \textstyle\sum_k (q\zeta_n^a)^k\Big) = N_{\mathbb{Q}(\zeta_n)/\mathbb{Q}}\big(x - \textstyle\sum_k (q\zeta_n)^k\big) \;\in\; \mathbb{Z}[x] \tag{23}$$

is a factor of $\det(x - L_{\varepsilon(q)})$, *of degree* $n-1$ *and with root-sum* $-(q + \cdots + q^{n-1})$.

(iv) χ_1 *is a character of* A_3 *and of* A_5, *and is not a character of* A_4 *or of* A_6. *At* $n = 7$ *the same test, run in* $\mathbb{F}_p S_7$ *for* $p = 999999001$, *finds* χ_1; *that case is evidence only. So the cyclotomic family exists exactly at the prime levels for every* n *in the available range, the two non-existences being theorems and the two existences being theorems proved by exact rational computation.*

Proof. (i) For odd n, $\mathrm{aug}(\varepsilon_{n,k}) = (-1)^{n-1} = 1 = \chi_0(\varepsilon_{n,k})$ for every k, and the $\varepsilon_{n,k}$ generate A_n. (ii) Lemma 6 gives $\varepsilon_{3,1} = c_3$ and $\varepsilon_{3,2} = c_3^{-1} = c_3^2$, so $A_3 = \mathbb{Q}[c_3]$, of dimension 3 because c_3 has order 3; the rest is the Chinese remainder theorem, and $d_{3,1} = d_{3,2} = 1$ makes sgn agree with aug on both generators. Formula (22) is the factorization of the characteristic polynomial over the three characters, and at $q = 2$ the three eigenvalues are 6, $2\zeta_3 + 4\zeta_3^2 = -3 - i\sqrt{3}$ and $2\zeta_3^2 + 4\zeta_3 = -3 + i\sqrt{3}$. (iii) $\chi_a^g(\varepsilon_{n,k}) = g(\zeta_n^{ak}) = \zeta_n^{gak} = \chi_{ga}(\varepsilon_{n,k})$, and χ_a^g is a character by Proposition 7(iii). The remaining assertions are Galois descent: $\sum_{a\neq 0} e_{\chi_a}$ is a Galois-fixed central idempotent of A_n, and the corresponding factor is a field because the χ_a, $a \neq 0$, are distinct and transitively permuted. (iv) Fix q and let $f \in \mathbb{Q}[x]$ be the minimal polynomial of the element $\varepsilon(q)$ of A_n. A character χ satisfies $0 = \chi\big(f(\varepsilon(q))\big) = f\big(\chi(\varepsilon(q))\big)$, so if χ_1 exists then $s_1 = \sum_k (q\zeta_n)^k$ is a root of f; since $N_n(s_1) = 0$ as well, $\gcd(f, N_n) \neq 1$. At $n = 4$ and $n = 6$ the Euclidean algorithm over $\mathbb{Q}$ gives $\gcd(f, N_n) = 1$, with $N_4 = x^2 + 8x + 52$ and $N_6 = x^2 + 2x + 1324$, so χ_1 does not exist; nothing beyond $f(\varepsilon(q)) = 0$ was used, so these two non-existences are unconditional. For the converse write each $\varepsilon_{n,k} = P_k\big(\varepsilon(q)\big)$, which is possible with $\deg P_k < \dim A_n$ exactly when $A_n = \mathbb{Q}[\varepsilon(q)]$; then $A_n \cong \mathbb{Q}[x]/(f)$ and $x \mapsto s_1$ defines a character of A_n, which is χ_1 if and only if $P_k(s_1) = \zeta_n^k$ for every k. At $n = 3, 5$ with $q = 2$ both halves hold: $\dim A_n = 3, 23 = \deg f$ (Remark 15(i)) and the identities $P_k(s_1) = \zeta_n^k$ hold in $\mathbb{Q}(\zeta_n)$, each step of both computations being exact rational arithmetic. The same computation at $n = 7$, carried out in $\mathbb{F}_p S_7$, also succeeds. □

Remark 14 (what is and is not known about the spectrum). For $n = 3$ the spectrum of $L_{\varepsilon(q)}$ is completely determined: the two roots $-3 \pm i\sqrt{3}$ that appeared as unidentified eigenvalues in earlier versions of this discussion are $\chi_1(\varepsilon(2))$ and $\chi_2(\varepsilon(2))$. For $n = 5$ Proposition 9 identifies six of the 23 characters — aug $= \chi_0$, sgn, and $\chi_1, \dots, \chi_4$ — and the latter four account for the factor $x^4 + 30x^3 + 580x^2 + 6840x + 36080$ of $\det(x - L_{\varepsilon(2)})$, whose root-sum $-30 = -(2 + 4 + 8 + 16)$ is the prediction of (23); the other four quartic factors of A_5 and the three rational characters besides aug and sgn remain unexplained. For $n = 4$ the factorization $\det(x - L_{\varepsilon(2)}) = (x-38)(x-14)(x-10)(x-6)(x+14)(x^2-8x+52)(x^2+24x+180)$ has the

two quadratic factors split by $\mathbb{Q}(i)$, but $i = \zeta_4$ is not $\chi_1(\varepsilon(2)) = 2i + 4i^2 + 8i^3 = -4 - 6i$, so $\mathbb{Q}(i)$ enters here for a reason the cyclotomic family does not explain. The near-miss is nonetheless exact up to sign: $N_4 = x^2 + 8x + 52$ has the roots $-4 \pm 6i$ while $N_4(-x) = x^2 - 8x + 52$ is the quadratic factor of $\det(x - L_{\varepsilon(2)})$, and at $n = 6$ the norm is $N_6 = x^2 + 2x + 1324$, with roots $-1 \pm 21i\sqrt{3}$, coprime to the degree-72 minimal polynomial of $\varepsilon(2)$; we have no explanation for either fact. No character beyond aug and sgn extends to $\mathbb{Q}S_n$, so all of these are genuinely new one-dimensional representations of a commutative algebra that is not a group algebra.

Remark 15 (structural evidence, positive and negative). Two probes for hidden structure behind the level rigidity of Corollary 2 are recorded here.

(i) *The level algebra.* Let $A_n \subseteq \mathbb{Q}S_n$ be the subalgebra generated by 1 and the $\varepsilon_{n,k}$ of Lemma 6. Computation in $\mathbb{F}_p S_n$, $p = 10^9 + 7$ (script `experiment_level_algebra.py`), cross-checked for $n \le 5$ by a second independent implementation working over $\mathbb{Q}$ in exact rational arithmetic, gives

$$\dim A_n = 3,\ 9,\ 23,\ 72 \qquad (n = 3, 4, 5, 6),$$

the pairwise commutativity of the $\varepsilon_{n,k}$, hence of A_n, and one further property that was not expected: the graded element $\varepsilon(q) = \sum_k \varepsilon_{n,k} q^k$ is a *cyclic* generator for $q \in \{2, 3, 5\}$, its minimal polynomial having degree $\dim A_n$ and being squarefree, so $A_n = \mathbb{Q}[\varepsilon(q)]$. For $n \le 5$ the two computations give the same dimension, so reduction modulo p is faithful on A_n and both properties — commutativity and cyclic generation — are theorems over $\mathbb{Q}$. At $n = 6$ commutativity survives on its own, because the commutators $[\varepsilon_{n,i}, \varepsilon_{n,j}]$ have coefficients bounded by $2n! < p$, so their vanishing modulo p is their vanishing over $\mathbb{Z}$; but reduction can only lower a rank, so the dimension is a lower bound, $\dim A_6 \ge 72$, and cyclic generation there remains evidence rather than proof. Étaleness, which at the time of the computation was a fourth empirical property, is now a theorem for every n: Proposition 7(ii).

The second script, `experiment_spectrum.py`, extends the range by one and reports the spectrum. Modulo $p = 999999001$, chosen so that $\zeta_n \in \mathbb{F}_p$ for $n \le 8$, the Krylov space of $\varepsilon(2)$ has dimension 223 at $n = 7$, so $\dim A_7 \ge 223$ against the centralizer bound $\dim A_7 \le 726$ below. The same script computes $\operatorname{tr}(L_{c_n^j})$, $0 \le j < n$, and hence the $\mathbb{Q}C_n$-multiplicities m_j of Proposition 7(iii):

$$(m_0, \dots, m_{n-1}) = (2, 2, 3, 2),\ (3, 5, 5, 5, 5),\ (10, 12, 12, 14, 12, 12),\ (25, 33, 33, 33, 33, 33, 33)$$

for $n = 4, 5, 6, 7$. Each is constant on the orbits of $(\mathbb{Z}/n)^\times$ as that proposition requires, and each sums to $\dim A_n$. The characters that take rational values on every $\varepsilon_{n,k}$ are, as vectors $\big(\chi(\varepsilon_{n,k})\big)_{k=1}^{n-1}$,

$$\begin{aligned}
n = 4:&\ (-1,-1,-1),\ (1,7,1),\ (1,1,1),\ (1,-1,1),\ (-1,5,-1);\\
n = 5:&\ (1,1,1,1),\ (1,21,21,1),\ (1,-3,-3,1);\\
n = 6:&\ (1,-1,1,-1,1),\ (-1,5,-1,5,-1),\ (1,1,1,1,1),\ (1,3,5,3,1),\\
&\ (-1,9,-1,9,-1),\ (1,11,-11,11,1),\ (1,51,161,51,1),\\
&\ (-1,-13,-25,-13,-1),\ (1,-3,-7,-3,1),\ (-1,-1,-1,-1,-1);\\
n = 7:&\ (1,1,1,1,1,1),\ (1,11,21,21,11,1),\ (1,31,1,1,31,1),\\
&\ (1,113,813,813,113,1),\ (1,-7,-3,-3,-7,1),\ (1,-5,1,1,-5,1),\\
&\ (1,-1,-3,-3,-1,1);
\end{aligned}$$

every one of them palindromic, as Proposition 7(ii) forces. The entries $(1,7,1) = (d_{4,k})_k$ and $(1,51,161,51,1) = (d_{6,k})_k$ and $(1,113,813,813,113,1) = (d_{7,k})_k$ are the sign character, and the constant vectors are the augmentation (all $(-1)^{n-1}$) and χ_0 (all 1), of Propositions 8 and 9. Wedderburn factors of the semisimple algebras are visible for $n \le 5$: $A_3 \cong \mathbb{Q} \times \mathbb{Q}(\zeta_3)$,

$A_4 \cong \mathbb{Q}^5 \times \mathbb{Q}(i)^2$, and $A_5 \cong \mathbb{Q}^3$ together with five quartic fields, one of them $\mathbb{Q}(\zeta_5)$, cut out by $x^4 + 30x^3 + 580x^2 + 6840x + 36080$ (Remark 14).

In this language Corollary 2 says that aug takes the value $(-1)^{n-1}$ on every generator, and Theorem 2 says that $F_n(q) = \mathrm{aug}\big(\varepsilon(q)\big)$: *the signed derangement enumerator is one eigenvalue of* $L_{\varepsilon(q)}$ *on* A_n, its sibling under Proposition 8 being the unsigned enumerator $d_n(q)$, and the remaining $\dim A_n - 2$ eigenvalues being those of Remark 14.

Lemma 6 has two unconditional consequences: A_n contains the group algebra $\mathbb{Q}\langle c_n\rangle \cong \mathbb{Q}C_n$, which for $n = 3$ is all of A_3, and $\varepsilon_{n,1}$ has minimal polynomial $x^n - 1$. If A_n is commutative — as it is for $n \le 7$ — then $\varepsilon_{n,1}$ is central in A_n, so $A_n \subseteq C_{\mathbb{Q}S_n}(c_n)$, whose dimension is the number of orbits of $\langle c_n\rangle$ acting on S_n by conjugation, $\frac{1}{n}\sum_{k=0}^{n-1}(n/g_k)^{g_k}\, g_k!$ with $g_k = \gcd(n,k)$; this bounds $\dim A_n$ by $4, 10, 28, 136, 726, 5100$ for $n = 3, \dots, 8$, against the computed $3, 9, 23, 72, 223$. Otherwise the data are negative: $\dim A_4 = 9$ is neither the order of a subgroup of S_4 (Lagrange) nor the dimension of the centre of $\mathbb{Q}S_4$ (which is 5), so A_4 is neither a group algebra nor central, and letting the $\varepsilon_{n,k}$ act on the subset space $\mathbb{F}_p^{2^{[n]}}$ yields an operator algebra of dimension $\binom{n}{2}$, so that natural model is not A_n either. We do not recognize the sequence $3, 9, 23, 72, 223$, whose growth rate stabilizes near 3.1; the probe structures the rigidity of Corollary 2 without yet explaining it. See Problem 5.

(ii) *Graded class sums*: the signed excedance class sums $S_\mu(q) = \sum_{\pi \in C_\mu} \mathrm{sgn}(\pi) q^{\mathrm{exc}(\pi)}$ are far from being central characters — already for the class of 4-cycles in S_4, $S_{(4)}(q) = -q - 4q^2 - q^3$ has three nonzero coefficients, whereas a central character is a scalar times a single class sum. Their total $\sum_\mu S_\mu(q) = (1-q)^{n-1}$ is just (3) re-indexed by conjugacy class. So the rigidity of Corollary 2 is not a centrality phenomenon; it is a parity/Euler-characteristic phenomenon whose structural counterparts, found after the first version of this note, are the conservation laws of the $q = -1$ section (the per-pair excedance balance of Lemma 1 and the reverse-complement pairing of Proposition 2) and, on the level algebra, the fact that aug is a character of A_n and rigidity is its value on the generators (Proposition 8).

Problem 5 (the level algebra)**.** Complete the description of the level algebra A_n begun in Lemma 7, Propositions 7, 8 and 9, and Remark 15(i). By Lemma 6 A_n contains $\mathbb{Q}C_n$; by Proposition 7 it is semisimple for every n, its character table is palindromic and Galois-symmetric, and it is commutative exactly when the coefficient symmetry $x_\pi = x_{\rho(\pi)}$ of Lemma 7 survives multiplication; by Proposition 8 the signed enumerator $F_n(q)$ of Theorem 2 and the unsigned enumerator $d_n(q)$ are two of its $\dim A_n$ eigenvalues. What is missing is (a) a proof of commutativity, equivalently of $\iota(x) = w_0 x w_0$ on A_n, and of cyclic generation by $\varepsilon(q)$, for every n; (b) the identification of the remaining $\dim A_n - 2$ characters — already complete for $n = 3$, where they are the level-multiplicative χ_1, χ_2 of Proposition 9(ii), and partial for $n = 5$ — together with a proof that χ_1 exists if and only if n is prime, which Proposition 9(iv) settles only for $n \le 6$; and (c) a closed form or recursion for $\dim A_n = 3, 9, 23, 72, 223, \dots$ Any one of the three would upgrade the rigidity of Corollary 2 from the Euler-characteristic accident of Problem 2 to a structural theorem, and would explain why the signed excedance levels of S_n behave like the cells of a space whose reduced homology is one-dimensional, concentrated at the survivor.

Acknowledgements

The question studied in this note was first discovered by the author in the spring semester of 2022, during the course *Linear Algebra II-B* at the University of the Chinese Academy of Sciences (UCAS); part of this work was carried out while the author was with the School of Artificial Intelligence, UCAS. The author warmly thanks the course instructor, Prof. Ming Fang (Academy of Mathematics and Systems Science, Chinese Academy of Sciences), and the teaching

assistant, Hongsheng Hu (now Associate Professor, Hunan University). This work was supported by the Beijing Natural Science Foundation (Grant No. QY23187).

Appendix A. Reproducibility

The accompanying script `verify.py` (run with `python3 verify.py`) recomputes, in exact integer and rational arithmetic, every enumerative claim of this note. Its 45 checks carry tags `[NN <label>]` in which `<label>` is the label of the statement checked; the same tags head the corresponding blocks of `verification_output.txt`, so every block of the report traces to a statement and conversely. The script needs Python 3 and, for the two-variable symbolic expansions of checks `[22]` and `[29]`, SymPy; everything else, including all of the group-algebra linear algebra of `[41]`–`[45]`, uses only the standard library.

Type A. `[01]` Theorem 1, for $1 \le n \le 7$ at several rational points (y, q) and at the removable singularity $q = 1$; `[02]` Theorem 2, coefficient by coefficient, $2 \le n \le 10$; `[03]` Corollary 2, $2 \le n \le 7$; `[05]` Corollary 7, $2 \le n \le 10$; `[14]` the weak-excedance specialization of Remark 1 (iv), $2 \le n \le 7$; `[15]` the symmetry in k of $(E_{n,k})$, $(O_{n,k})$ and $(d_{n,k})$ of Remark 3, $2 \le n \le 8$; `[17]` the obstruction recorded in Problem 2 (the greedy exc-neutral adjacent swap is not an involution: $n \le 6$ on S_n, $n \le 5$ on $\mathcal{D}_n$); `[18]`, `[19]`, `[21]` the $q = -1$ section: Proposition 2 for $n = 3, 5$, the split of Proposition 3 for $n = 2, 4, 6, 8$ together with the identification of the reverse-complementary derangements with $\mathcal{D}(B_m)$ and their counts $1, 5, 29, 233$ [11], and Lemmas 1–2; `[22]` Proposition 1, $n \le 6$; `[28]` Corollary 6, $n \le 6$; `[29]` Corollary 1, $n \le 6$; `[30]` the triple enumerator of Remark 6, $n \le 6$; `[34]`, `[35]` the two probes of Remark 15 (the subalgebra of $\mathbb{Q}S_n$ generated by the identity and the level elements, $n = 3, 4$; and the signed excedance class sums of S_4); `[40]` Lemma 6, $2 \le n \le 9$ (the unique derangements with one and with $n-1$ excedances, their sign, and $\varepsilon^n_{n,1} = 1$ in $\mathbb{Q}S_n$); `[41]` Lemma 7, $2 \le n \le 8$; `[42]` Proposition 7(i), $n = 3, 4, 5$: commutativity and ρ-invariance on A_n, and a non-commuting ρ-fixed pair outside it; `[43]` Proposition 7(ii),(iii), $n = 3, 4, 5$: cyclicity of $\varepsilon(q)$, the trace form $(\iota(x)y)_e$ and its positive definiteness, squarefreeness of f, $\iota(\varepsilon(q)) = q^n\varepsilon(1/q)$, and the cyclotomic factorization of $\det(x - L_{c_n})$ on A_n, cross-checked against the traces $\mathrm{tr}(L_{c_n^k}) = \sum_d e_d\, c_d(k)$; `[44]` Proposition 8, $2 \le n \le 7$; `[45]` Proposition 9, $n = 3, 4, 5, 6$ at $q = 2$: the norm N_n and its closed forms for $n = 3, 4$, (22) at $q = 2, 3, 5$, the $\gcd(f, N_n)$ test for χ_1, and, where the test succeeds, the specialization $\varepsilon_{n,k} \mapsto \zeta_n^k$ carried out in $\mathbb{Q}(\zeta_n) = \mathbb{Q}[t]/(\Phi_n)$.

Type B. `[04]` the two-variable identity of Theorem 3, $\Phi_n(x, q) = (x+1)(1-q)^{n-1}$ coefficient by coefficient together with the standard-character companion (12), $n \le 6$; `[06]`, `[07]`, `[12]` Proposition 4 and Theorem 3(i)–(ii), $n \le 8$; `[08]` Theorem 4, $n \le 8$; `[09]`, `[10]` Proposition 5, by summation over the half for $2 \le n \le 8$ and by the explicit involution $t = (1\,2)$ exhaustively for $n \le 6$; `[11]` Theorem 5, $n \le 8$; `[13]` the type-D identity of Remark 9, $n \le 8$; `[26]` Lemma 4, exhaustively for $n \le 5$; `[27]` the pairing-graph perfect matching on the vanishing half supporting Problem 1, by a Tutte-matrix rank test modulo a prime, $n \le 5$; `[33]` the inv-weighted identity of Problem 4, $n \le 6$.

The graded Alon–Tarsi polynomial. `[20]` $\mathrm{AT}_n^{\mathrm{cyc}}(q)$ of (15) by distance to the cyclic square, $n = 3, 4$; `[23]` the determinant-sum formula (16) against the sparsity grading (17), $n \le 4$; `[24]` Corollary 8, $n = 2, 4$; `[25]` the layer-wise vanishing of Theorem 6 at $n = 5$, layers $5, 6$; `[31]` Proposition 6, $2 \le n \le 40$; `[32]` Corollary 9, $n \le 4$; `[36]` the normalization of (16) against the Latin-square definition, $n = 2, 3, 4$.

Data and timings. `[37]`, `[38]` the data of Tables 1 and 2; `[39]` the even/odd pairs per excedance level quoted in the introduction; `[16]` benchmark timings comparing the brute-force computation of $F_n(q)$ with the closed form (8) ($n \le 9$; brute force dominates as $O(n!\, n^2)$).

The full output is stored in `verification_output.txt`; the run takes about three minutes on one core, the Krylov space of $\varepsilon(2)$ in $\mathbb{Q}S_6$ dominating it. The second script, `experiment_survivors.py`, reproduces the two computational observations behind Problem 2: which derangements can serve as the unique survivor of a prescribed excedance level, and the existence of perfect twist-flipping pairings on the type-B vanishing half. The third, `experiment_level_algebra.py`, reproduces the structural data of Remark 15(i), Lemma 6 and Problem 5: for $n \le 6$, by linear algebra over $\mathbb{F}_p$, $p = 10^9 + 7$, the dimension of the level algebra A_n, the commutativity of the level elements, the rank of the trace form and the minimal polynomials of $\varepsilon_{n,1}$ and $\varepsilon(q)$, and the centralizer bound $\dim A_n \le \dim C_{\mathbb{Q}S_n}(c_n)$; for $n \le 5$ the same dimensions in exact rational arithmetic, by a second independent implementation of the group algebra; and for $n \le 9$, by brute force, the extreme levels of Lemma 6. The fourth, `experiment_spectrum.py`, reproduces the spectral data of Remark 15(i), Proposition 9(iv) and Remark 14: exactly over $\mathbb{Q}$ for $n \le 5$ and modulo $p = 999999001$ for $4 \le n \le 7$, the factorization of the minimal polynomial of $\varepsilon(q)$ into Wedderburn factors, the multiplicities m_j, the rational characters as palindromic value vectors, and the divisibility test for the cyclotomic candidates χ_a.

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Department of Automation, Tsinghua University, School of Artificial Intelligence, University of Chinese Academy of Sciences